

Original theta Hankel certification

Complete proof sources

This separate support reader contains the complete original counterfactual workbench note and the complete independent proofs of its source and packet maps, the certified finite Hankel calculation, and its positive theta tails. Every mathematical body appears in full. Original source files and exact conversion records accompany the reader.

The original workbench's numerical probes retain their stated exploratory scope. The later certification proofs concern the displayed original finite matrices and the exact recorded intervals. Their finite conclusion and every source qualification remain attached to those objects.

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The complete original counterfactual workbench note

Counterfactual RH search on the original split-supported theta input

Scope

The preceding Deligne-bound audit is not a no-go theorem for geometry over F1, for the monoidal base tau, or for a future purity construction using the full arithmetic input. Its stated conclusion concerns a particular inference from an auxiliary exponential family's Frobenius purity to a bound for the original theta metric. Its examples do not assert that their chosen polynomials divide the original function $g = 2\xi$.

This continuation takes the counterexample search as an independent research objective. It proves a finite-sign certificate for a counterexample to RH, directly from derivatives of the original theta function. It also proves that an actual counterexample would eventually supply such a certificate. The finite calculation reported below found no negative witness. No counterexample to RH is claimed.

The power-sum/moment method belongs to established zero-positivity methods; see Zhang (2015). The derivation below gives all its maps, signs, and constants for the present source, rather than claiming a new general RH criterion.

1. Original input and support

Retain

$$G(R) = \{\tau\} \sqcup R^\bullet, \quad e_R = 0_R^\bullet, \quad p_R(\tau) = 0, \quad p_R(r^\bullet) = r,$$

and the coefficient square

$$\begin{array}{ccc} G(\mathbb{Z}) & \longrightarrow & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \longrightarrow & \mathbb{C}. \end{array}$$

The support observation sends τ to Boolean zero and every represented coefficient to Boolean one. On the original support-labelled coefficient fibres, a linear map lifts as $(\ell, v) \mapsto (\ell, fv)$, with external absence fixed. A killed coefficient therefore reaches $(\ell, 0)$, not external absence. No conclusion about the sign of an arithmetic functional follows just from that support value.

Use the unchanged source

$$\Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2},$$

$$f_0 = \Theta\phi_*, \quad \mathcal{M}f_0(s) = g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

These are the arithmetic-input equations (A1)–(A3), also identified with the specified Hochschild trace in HOCHSCHILD_COMPARISON.md, equations (H4)–(H10). Define the explicit logarithmic-coordinate function

$$\psi(y) = e^{y/2} f_0(e^y). \tag{1}$$

Mellin substitution gives

$$g(1/2 + iz) = \int_{\mathbb{R}} \psi(y) e^{izy} dy. \tag{2}$$

The original reflection $f_0(x) = x^{-1}f_0(1/x)$ gives $\psi(-y) = \psi(y)$. This reflection also follows by applying the source Fourier identity to ϕ_* : Fourier sends D to $1 - D$, so it fixes $(D^2 - D)e^{-\pi x^2} = \phi_*$.

For $x \geq 1$, every summand in $f_0(x)$ is positive, since

$$4\pi^2 n^4 x^4 - 6\pi n^2 x^2 = 2\pi n^2 x^2 (2\pi n^2 x^2 - 3) > 0.$$

Reflection proves positivity for all $x > 0$. The Gaussian series also gives a constant C such that

$$0 < \psi(y) \leq C \exp(9|y|/2 - \pi e^{2|y|}). \quad (3)$$

For $y \geq 0$, bound the convergent sum of $n^4 e^{-\pi(n^2-1)e^{2y}}$ by its value at $y = 0$; reflection supplies the other half-line. Thus all differentiations and all moments below converge absolutely.

2. Coefficients with the original mass retained

Put

$$M_{2j} = \int_{\mathbb{R}} y^{2j} \psi(y) dy = g^{(2j)}(1/2), \quad a_j = \frac{(-1)^j M_{2j}}{(2j)!}. \quad (4)$$

Define the entire function F by its specified coefficients:

$$F(w) = \sum_{j \geq 0} a_j w^j, \quad F(z^2) = g(1/2 + iz), \quad a_0 = g(1/2) > 0. \quad (5)$$

No square-root branch is part of the definition, and a_0 is not assigned the value one. The map $z \mapsto z^2$ identifies the two reflected coordinates only at this scalar-function observation; the original packet modules are retained in Section 5.

Define b_n by the analytic logarithmic derivative near zero:

$$-\frac{F'(w)}{F(w)} = \sum_{n \geq 0} b_n w^n. \quad (6)$$

The complete finite coefficient map is

$$b_n = \frac{-(n+1)a_{n+1} - \sum_{j=1}^n a_j b_{n-j}}{a_0}. \quad (7)$$

It follows by multiplying (6) by F and extracting the coefficient of w^n . For example,

$$b_0 = -\frac{a_1}{a_0}, \quad b_1 = \frac{a_1^2}{a_0^2} - \frac{2a_2}{a_0}, \quad b_2 = -\frac{a_1^3}{a_0^3} + \frac{3a_1 a_2}{a_0^2} - \frac{3a_3}{a_0}. \quad (8)$$

This operation need not preserve positivity of arbitrary coefficient sequences. The matrices used below are not positive by their definition.

3. A finite certificate contradicting RH

Let α_j be the distinct zeros of F , with their multiplicities m_j , and put $\beta_j = \alpha_j^{-1}$. These symbols are used to prove the test, not supplied to its numerical implementation.

Estimate (3), followed by $t = e^{2|y|}$, bounds the maximum modulus of g by an exponential of $O(R \log R)$. The maximum modulus of F at radius R is consequently bounded by an exponential of $O(\sqrt{R} \log R)$. Its order is at most $1/2 < 1$. Genus-zero Hadamard factorization therefore gives

$$F(w) = a_0 \prod_j (1 - \beta_j w)^{m_j}, \quad \sum_j m_j |\beta_j| < \infty. \quad (9)$$

There is no nonconstant exponential factor of integer degree at this order. Differentiating locally uniformly away from zeros proves

$$b_n = \sum_j m_j \beta_j^{n+1}. \quad (10)$$

The exact coordinate of a zeta zero $\rho = 1/2 + \delta + i\gamma$ is

$$z = -i(\rho - 1/2) = \gamma - i\delta, \quad \alpha = z^2 = \gamma^2 - \delta^2 - 2i\gamma\delta,$$

$$\boxed{\beta = -\frac{1}{(\rho - 1/2)^2} = \frac{\gamma^2 - \delta^2 + 2i\gamma\delta}{(\gamma^2 + \delta^2)^2}.} \quad (11)$$

The two reflected roots $\rho, 1 - \rho$ give one zero α of the same multiplicity in F . Conjugation gives the conjugate α and β .

For every real s , (1)–(3) give $g(s) = \int \psi(y)e^{(s-1/2)y}dy > 0$. Hence F has no nonpositive real zeros: $F(-v^2) = g(1/2 - v) > 0$. Thus RH is equivalent to every β_j being positive real.

For $d \geq 0$, define the actual Hankel coefficient matrix

$$\mathbf{H}_d = (b_{r+s})_{0 \leq r, s \leq d}. \quad (12)$$

For $P(X) = \sum_{r=0}^d c_r X^r$ with real coefficients, the exact value is

$$\boxed{c^T \mathbf{H}_d c = \sum_j m_j \beta_j P(\beta_j)^2.} \quad (13)$$

Absolute convergence follows from (9) and boundedness of the polynomial on the compact set of reciprocal zeros and zero. Conjugate pairs make (13) real.

Under RH, each term is nonnegative. Therefore

$$\boxed{c^T \mathbf{H}_d c < 0 \implies \text{RH is false.}} \quad (14)$$

The certificate may use $c \in \mathbb{Q}^{d+1}$: a strict negative value persists under sufficiently small rational coefficient perturbations. With validated enclosures $b_n \in [\underline{b}_n, \bar{b}_n]$, exact rational interval arithmetic on $\sum c_r c_s b_{r+s}$ is enough to certify that its upper endpoint is negative.

A second necessary family is $\mathbf{H}_d^{(1)} = (b_{r+s+1})$. Under RH its quadratic form is $\sum_j m_j \beta_j^2 P(\beta_j)^2 \geq 0$. We tested both, though the unshifted family alone is complete by the next construction.

4. Work through the off-line counterfactual: construct its negative direction

Assume for this argument that RH is false. Equation (11) supplies a nonreal reciprocal zero β and its conjugate, each of multiplicity m . They are nonreal because g has no real zeros, so both γ and δ are nonzero.

Put $r = |\beta|/2$. There are finitely many distinct reciprocal zeros of modulus at least r . Remove $\beta, \bar{\beta}$ from that finite set and call the remaining set $\mathcal{A}$. It is closed under conjugation. Define the real polynomial

$$B(X) = \prod_{\gamma \in \mathcal{A}} (X - \gamma). \quad (15)$$

Choose $c \in \mathbb{C}$ satisfying $\beta c^2 = -1$; either of the two choices works. For an integer $N \geq 0$, put

$$w_N = \frac{c}{\beta^N B(\beta)}, \quad v_N = \frac{\operatorname{Im} w_N}{\operatorname{Im} \beta}, \quad u_N = \operatorname{Re} w_N - \operatorname{Re} \beta v_N,$$

$$\boxed{P_N(X) = X^N B(X)(\nu_N + v_N X).} \quad (16)$$

The coefficients are real. At the two target points,

$$P_N(\beta) = c, \quad P_N(\bar{\beta}) = \bar{c}.$$

At every other point of modulus at least r , P_N vanishes. The two target contributions to (13) are exactly

$$m\beta c^2 + m\bar{\beta}\bar{c}^2 = -2m. \quad (17)$$

Here is an explicit bound on everything else. Let

$$B_0 = \prod_{\gamma \in \mathcal{A}} (r + |\gamma|), \quad C = \frac{|c|B_0}{|B(\beta)|} \left(1 + \frac{|\operatorname{Re} \beta| + r}{|\operatorname{Im} \beta|} \right),$$

$$T = \sum_{|\beta_j| < r} m_j |\beta_j| < \infty.$$

For $|X| \leq r$, the coefficient definitions imply

$$|P_N(X)| \leq C \left(\frac{r}{|\beta|} \right)^N = C^2 2^{-N}.$$

Consequently the complete, real quadratic form satisfies

$$\boxed{\mathcal{L}(P_N^2) := \sum_j m_j \beta_j P_N(\beta_j)^2 \leq -2m + C^2 T 4^{-N}.} \quad (18)$$

For all sufficiently large N the right side is strictly negative. Every omitted reciprocal zero is included in the tail estimate, and every multiplicity is retained. This proves

$$\boxed{\text{RH} \iff \mathbf{H}_d \succeq 0 \text{ for every } d \geq 0.} \quad (19)$$

It also explains the limitations of a finite search. The degree needed in (16) depends on the location of the off-line pair, the other larger reciprocal zeros, and the tail. The proof does not say that a counterexample must be visible in a small matrix. The polynomial (16) is a counterfactual construction; no actual off-line pair has been used to instantiate it.

5. Connect the witness to the original finite packet, including its jets

For a finite conjugation- and reflection-stable packet of actual zeros, retain the original inclusion

$$\sigma_h : E_h = \mathbb{C}[s]/(h) \hookrightarrow Q, \quad D_Q \sigma_h = \sigma_h A_h.$$

The original arithmetic input constructs this through the kernel of multiplication by $j_h g$. An arbitrary polynomial example is not automatically such a packet.

Because $g(1/2) > 0$, the operator $A_h - \frac{1}{2}I$ is invertible on every actual packet. Define the specified rational functional calculus

$$T_h = -(A_h - \tfrac{1}{2}I)^{-2}, \quad \mathbb{C}[X] \longrightarrow \operatorname{End}_{\mathbb{C}}(E_h), \quad P \longmapsto P(T_h). \quad (20)$$

On a full local block $A_h = \rho I + N_\rho$, $N_\rho^m = 0$, the matrix is exactly

$$\boxed{T_h = \sum_{j=0}^{m-1} (-1)^{j+1} (j+1) (\rho - 1/2)^{-j-2} N_\rho^j.} \quad (21)$$

The coefficient of N_ρ is $2(\rho - 1/2)^{-3} \neq 0$, so this local coordinate change retains the nilpotent length. The two reflected primary modules can have the same T_h eigenvalue, but they remain separate in E_h .

The packet contribution to (13) is

$$\boxed{\mathcal{L}_h(P^2) = \tfrac{1}{2} \operatorname{Tr}_{E_h}(T_h P(T_h)^2).} \quad (22)$$

The factor $1/2$ counts each reflected pair once, matching (5). The trace records algebraic multiplicities, while the full operators and their nilpotent coefficients remain upstream of the trace map. Absolute convergence in (9) makes the full functional the limit over packets exhausting the actual zero set.

The existing critical-line comparison satisfies

$$\ker(\Gamma\sigma_h) = E_{h,\text{off}}.$$

Equation (22) is evaluated on the source packet before applying that map. An off-critical class maps to $(\ell, 0)$ under the split lift of Γ , but is not deleted from the source on which the trace and negative witness are evaluated. This is why the search does not obtain a false positive or a false proof by restricting to the critical-only target.

6. The performed search uses the original theta seed, not zero ordinates

The numerical calculation uses (4), in the explicit form

$$M_{2j} = 4 \sum_{n \geq 1} \int_0^\infty y^{2j} (4\pi^2 n^4 e^{9y/2} - 6\pi n^2 e^{5y/2}) e^{-\pi n^2 e^{2y}} dy. \quad (23)$$

The factor four consists of the source theta factor two and the reflection factor two. The implementation computes $M_0, \dots, M_{64}$, forms (7), and tests both matrix families through dimension 16 by an unpivoted LDL^T calculation in their original monomial coordinates. No rescaling or preconditioning is used. No zero ordinate is supplied.

Two runs were completed:

Working decimal precision	Gauss–Legendre nodes	Retained theta terms	y interval	Outcome
120	384	12	[0,4]	All 16 pivots positive numerically, in both families
180	768	14	[0,4]	Same outcome

Selected values are

$$a_0 = 0.9942415563766282198255474793707954396 \dots,$$

$$b_0 = 0.0231049931154189707889338104303390140 \dots,$$

$$b_1 = 0.00003717259928526968616486626247174057845 \dots,$$

$$b_2 = 0.0000001441739314009732796953815560948209 \dots,$$

$$\det H_1 = b_0 b_2 - b_1^2 = 1.949335554819142214739878219172139506 \times 10^{-9} \dots$$

The largest relative difference between the two runs' stored pivot values is below 3×10^{-79} . Those stored values contain 80 significant digits, so the comparison does not support a stronger precision claim. The scripts report the input choices and complete pivot sequences.

These are exploratory multiple-precision quadratures, not interval-certified integrations. A failed attempt to install an interval-arithmetic backend did not produce an interval certificate; only the documented mpmath computations are reported. Positive numerical pivots do not prove the corresponding signs, and a negative numerical pivot would require certified analytic enclosures before it could count as a disproof. No negative numerical pivot was found.

7. Finite verification and the next decisive object

Six standard-library exact rational checks passed in normal and optimized Python. They verify the logarithmic recurrence, positive-node identity, an explicit nonreal-pair negative control, and a rational interval evaluator that cannot call an interval containing zero strictly negative. The negative control is expressly an auxiliary polynomial, not an actual zeta packet.

The interval evaluator’s contract is limited: it checks the quadratic form from supplied valid intervals. Establishing that those intervals enclose the actual b_n remains an analytic task. No Lean certificate is claimed.

The decisive counterexample object in this branch is now a finite tuple

$$(d, c_0, \dots, c_d, [b_0, \bar{b}_0], \dots, [b_{2d}, \bar{b}_{2d}])$$

with rational c_r , rigorously justified coefficient enclosures from the original theta input, and a strictly negative upper bound for $\sum_{r,s} c_r c_s b_{r+s}$. Equation (14) proves exactly what that tuple would establish. Equation (18) proves that every actual off-line counterexample has a finite witness of this type at some degree.

The calculation performed here did not find that tuple. The failure of the previous auxiliary-purity inference is not used as evidence that one exists. Geometry over the original absolute base remains a separate possible source of further constraints; this note establishes no impossibility theorem about it.

Source and execution record

- Original arithmetic source: supplied `arithmetic_input.tex`, equations (A1)–(A10), including the actual multiplication-by- $j_h g$ kernel sequence.
- Original support reconstruction and internal homology: supplied SplitZero reconstruction record, Sections 1–4. The source-labelled zero and the external absent value are kept distinct throughout.
- Original critical comparison: supplied `HOCHSCHILD_COMPARISON.md`, equations (H16)–(H25). The visible text supplies the exact off-critical kernel and critical-jet retractions.
- Prior scope: supplied `Tau_Deligne_Bound_Audit/NOTE.tex`, “Result and scope”. It explicitly does not refute the split-zero construction or establish falsity of the arithmetic estimate for 2ξ .
- Classical context: R. Zhang, *On Power Sums of Positive Numbers*, arXiv:1510.03420 (2015). Its abstract documents moment-theoretic positivity criteria for genus-zero entire functions and applications to RH. The concrete criterion and separation proof above are proved here, not attributed as a verbatim theorem from an uninspected full text.
- Classical background: Hadamard factorization for entire functions of order less than one; the functional equation and completed-zeta notation are also recorded in NIST DLMF Section 25.4. The source Gaussian estimate above supplies the order bound needed for this application.
- Computation: mpmath 1.3.0. Results are in `probe_120dps.json` and `probe_180dps.json`; reproduction script is `search_theta_moments.py`. Exact finite checks and the interval-input verifier are in `exact_witness_checks.py`.

No GitHub file, branch, comment, or remote publication was changed in this continuation.

Independent proof and full packet transport

Original theta Hankel witness: independent proof and full packet transport

This is an additive mathematical review of the complete supplied NOTE.md, equations (1)–(23), and its complete LaTeX counterpart. Both original files remain byte-for-byte unchanged. The proof of the Hankel criterion and the infinite-tail separator is accepted after the explicit symbol repair below. This proof supplies explicit growth constants, the inverse local coordinate, the complete finite algebra map, and the original theta-boundary representative of that map. The source note's numerical results retain their exploratory status.

The source defines $u_N = \Re w_N - (\Re \beta)v_N$, but its boxed equation (16) uses the undefined symbol ν_N . The exact repair is the identity

$$\nu_N := u_N = \Re w_N - (\Re \beta)v_N.$$

This occurs at NOTE.md lines 212/216 and NOTE.tex lines 350/354. The user's pasted narrative already uses u_N in both places. Equation (WBR16) below writes that complete definition and the polynomial using the same u_N . No value or coefficient is changed.

WBR1. Source and scalar factors

Let

$$\begin{aligned} D &= -x\partial_x, \quad \hat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \\ V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \ (b \in \mathbb{Z}, j \geq 0)\}, \\ \Theta : V &\rightarrow \mathcal{B}, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \end{aligned} \tag{WBR1}$$

The quotient is the original $Q = \mathcal{B}/\Theta V$, with its quotient topology. Its split coefficient set is $G(R) = \{\tau\} \sqcup R^\bullet$, with represented zero $e_R = 0_R^\bullet$. A map on a coefficient fibre sends (ℓ, v) to (ℓ, fv) and fixes external absence τ . Thus $(\ell, 0)$ keeps the source label.

Put

$$q_0 = e^{-\pi x^2}, \quad \phi_* = (D^2 - D)q_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad f_0 = \Theta\phi_*. \tag{WBR2}$$

Direct differentiation gives $Dq_0 = 2\pi x^2 q_0$ and $D^2 q_0 = (4\pi^2 x^4 - 4\pi x^2)q_0$. The function ϕ_* is even and vanishes at zero. Integration by parts gives $\int Df = \int f$ on Schwartz functions, so $\int \phi_* = 0$. Hence $\phi_* \in V$. Fourier differentiation and integration by parts give $\mathcal{F}D = (1 - D)\mathcal{F}$. Since $\mathcal{F}q_0 = q_0$, this proves $\mathcal{F}\phi_* = \phi_*$. Poisson summation, with zero origin terms on both sides, gives

$$f_0(x) = x^{-1} f_0(1/x). \tag{WBR3}$$

For $x \geq 1$, the series and its derivatives decay faster than every power. Reflection proves the same assertion at zero, so $f_0 \in \mathcal{B}$.

For $\Re s > 1$, absolute convergence allows termwise integration:

$$\begin{aligned} \mathcal{M}f_0(s) &= 2\zeta(s) \int_0^\infty (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2} x^{s-1} dx \\ &= 2\zeta(s) \pi^{-s/2} \{2\Gamma(s/2 + 2) - 3\Gamma(s/2 + 1)\} \\ &= s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s) = 2\xi(s). \end{aligned} \tag{WBR4}$$

The two-ended decay makes the Mellin transform entire. Thus this identity also supplies the entire continuation of the completed zeta expression. Write $g = 2\xi$, retaining its actual value at $1/2$. The conventional factor in ξ is given by DLMF 25.4.4.

WBR2. Positive density and explicit growth

Define

$$\psi(y) = e^{y/2} f_0(e^y), \quad C_\theta = 8\pi^2 \sum_{n=1}^{\infty} n^4 e^{-\pi(n^2-1)}. \quad (\text{WBR5})$$

The constant is finite. Reflection gives $\psi(-y) = \psi(y)$. For $y \geq 0$, each theta summand is strictly positive because $2\pi n^2 e^{2y} > 3$. Dropping its negative term only for an upper bound, while retaining the complete summand in the definition, yields

$$0 < \psi(y) \leq 8\pi^2 e^{9y/2} \sum_{n \geq 1} n^4 e^{-\pi n^2 e^{2y}} \leq C_\theta e^{9y/2 - \pi e^{2y}}. \quad (\text{WBR6})$$

Reflection gives the same estimate with $|y|$ on the whole line. For any compact set of s , this dominates every $|y|^j \psi(y) e^{(s-1/2)y}$ by an integrable function. Therefore

$$g(s) = \int_{\mathbb{R}} \psi(y) e^{(s-1/2)y} dy, \quad M_{2j} = \int_{\mathbb{R}} y^{2j} \psi(y) dy = g^{(2j)}(1/2). \quad (\text{WBR7})$$

In particular $g(s) > 0$ for every real s , without numerical input.

The entire function $G(z) = g(1/2 + iz)$ is even. Its Taylor series gives

$$G(z) = \sum_{j \geq 0} \frac{(-1)^j M_{2j}}{(2j)!} z^{2j} = F(z^2), \quad F(w) = \sum_{j \geq 0} a_j w^j, \quad a_j = \frac{(-1)^j M_{2j}}{(2j)!}. \quad (\text{WBR8})$$

The map $z \mapsto z^2$ sends the circle of radius $\sqrt{R}$ onto the circle of radius R , so the convergent Taylor series proves that F is entire and $M_F(R) = M_G(\sqrt{R})$. No square-root branch is chosen. Its unchanged mass is $F(0) = a_0 = g(1/2) > 0$.

Set $a(R) = 9/4 + \sqrt{R}/2$. Substituting $t = e^{2y}$, with $dy = dt/(2t)$, into the majorant gives

$$\begin{aligned} M_F(R) &\leq 2C_\theta \int_0^\infty e^{(9/2 + \sqrt{R})y - \pi e^{2y}} dy \\ &= C_\theta \int_1^\infty t^{a(R)-1} e^{-\pi t} dt \\ &\leq C_\theta \pi^{-a(R)} \Gamma(a(R)) \\ &\leq 2C_\theta \pi^{-a(R)} \left(\frac{2(a(R)-1)}{e} \right)^{a(R)-1}. \end{aligned} \quad (\text{WBR9})$$

For the last inequality, maximize $t^{a-1} e^{-t/2}$ at $2(a-1)$ and integrate the remaining $e^{-t/2}$. Here $a(R) > 1$. Thus $\log M_F(R) = O(\sqrt{R} \log R)$, and the order is at most $1/2$.

WBR3. Reciprocal zeros and coefficient map

Use the standard genus-zero case of Hadamard factorization in this precise form: an entire function of finite order strictly below one, nonzero at zero, is its value at zero times the product $1 - w/\alpha$ over its zeros with multiplicity; the reciprocal absolute zero values are summable, and no nonconstant exponential factor occurs. WBR2 verifies its hypotheses for the actual F .

Let α_j be its distinct zeros, m_j their orders, and $\beta_j = 1/\alpha_j$. Then

$$F(w) = a_0 \prod_j (1 - \beta_j w)^{m_j}, \quad S_\beta = \sum_j m_j |\beta_j| < \infty. \quad (\text{WBR10})$$

The reciprocal zeros are nonzero and have only zero as a possible accumulation point. On a disk where all $|\beta_j w| < 1/2$, the logarithmic derivative is dominated by $2 \sum m_j |\beta_j|$. Geometric expansion and absolute convergence give

$$-\frac{F'(w)}{F(w)} = \sum_j \frac{m_j \beta_j}{1 - \beta_j w} = \sum_{n \geq 0} b_n w^n, \quad b_n = \sum_j m_j \beta_j^{n+1}. \quad (\text{WBR11})$$

Multiplying by F and equating coefficients, with a_0 retained, gives

$$b_n = \frac{-(n+1)a_{n+1} - \sum_{j=1}^n a_j b_{n-j}}{a_0}. \quad (\text{WBR12})$$

These b_n are real because all a_j are real.

At a zero $\rho = 1/2 + \delta + i\gamma$, the exact coordinates are

$$z = -i(\rho - 1/2) = \gamma - i\delta, \quad \alpha = z^2, \quad \beta = -\frac{1}{(\rho - 1/2)^2} = \frac{\gamma^2 - \delta^2 + 2i\gamma\delta}{(\gamma^2 + \delta^2)^2}. \quad (\text{WBR13})$$

Since $g(1/2) > 0$, $z \neq 0$, so $z \mapsto z^2$ has nonzero derivative and preserves the local zero order. The two reflected points $\rho, 1 - \rho$ supply one zero α of that order. Conjugation preserves the orders and sends α, β to their conjugates. For real v , $F(-v^2) = g(1/2 - v) > 0$. Thus no nonpositive real zero occurs, and RH is equivalent to every β_j being positive real. An off-line zero has $\delta \neq 0$, while positivity of g on the real axis gives $\gamma \neq 0$; its reciprocal coordinate is therefore nonreal by (WBR13).

WBR4. Finite criterion and full infinite-tail proof

For $P(X) = \sum_{r=0}^d c_r X^r$, with real coefficients, put

$$H_d = (b_{r+s})_{0 \leq r, s \leq d}, \quad \mathcal{L}(P^2) = c^T H_d c = \sum_j m_j \beta_j P(\beta_j)^2. \quad (\text{WBR14})$$

The reciprocal zeros together with zero form a compact set. Thus boundedness of P and (WBR10) prove absolute convergence. Conjugate terms make the sum real. Under RH its terms are nonnegative, proving $H_d \succeq 0$. The square retained in this formula is the algebraic $P(\beta)^2$.

For the converse, choose a nonreal reciprocal zero β of order m , and its conjugate. Set $r = |\beta|/2$. Among the finitely many distinct reciprocal zeros of modulus at least r , remove these two and call the remaining set $\mathcal{A}$. Define

$$B(X) = \prod_{\gamma \in \mathcal{A}} (X - \gamma), \quad B_0 = \prod_{\gamma \in \mathcal{A}} (r + |\gamma|). \quad (\text{WBR15})$$

The empty products are one. Conjugation permutes $\mathcal{A}$, so B is real and $B(\beta) \neq 0$. Choose either root c of $\beta c^2 = -1$. For an integer $N \geq 0$, put

$$w_N = \frac{c}{\beta^N B(\beta)}, \quad v_N = \frac{\Im w_N}{\Im \beta}, \quad u_N = \Re w_N - (\Re \beta) v_N, \quad P_N(X) = X^N B(X) (u_N + v_N X). \quad (\text{WBR16})$$

The coefficients are real, $u_N + v_N \beta = w_N$, and hence $P_N(\beta) = c$ and $P_N(\bar{\beta}) = \bar{c}$. The selected contributions total $-2m$. Every other point of modulus at least r contributes zero, regardless of its multiplicity.

The remaining infinite tail has the explicit constants

$$C = \frac{|c| B_0}{|B(\beta)|} \left(1 + \frac{|\Re \beta| + r}{|\Im \beta|} \right), \quad T = \sum_{|\beta_j| < r} m_j |\beta_j|. \quad (\text{WBR17})$$

Indeed $|v_N| \leq |w_N|/|\Im \beta|$ and $|u_N| \leq |w_N|(1 + |\Re \beta|/|\Im \beta|)$. For $|X| \leq r$, $|B(X)| \leq B_0$, giving $|P_N(X)| \leq C 2^{-N}$. Therefore

$$\left| \sum_{|\beta_j| < r} m_j \beta_j P_N(\beta_j)^2 \right| \leq C^2 T 4^{-N}, \quad \mathcal{L}(P_N^2) \leq -2m + C^2 T 4^{-N}. \quad (\text{WBR18})$$

The tail is real by conjugation, which justifies the ordered inequality. If $T = 0$, take $N = 0$. If $T > 0$, the explicit choice

$$N = \max \left(0, 1 + \left\lfloor \log_4 \left(\frac{C^2 T}{2m} \right) \right\rfloor \right) \quad (\text{WBR19})$$

satisfies $4^N > C^2 T / (2m)$. This yields a finite negative direction of degree at most $N + |\mathcal{A}| + 1$, proving

$$\text{RH} \iff H_d \succeq 0 \quad \text{for every } d \geq 0. \quad (\text{WBR20})$$

Every actual zero and every multiplicity is included in this proof. The degree depends on the specified pair and tail; no uniform degree or actual off-line zero is thereby supplied.

For a strict negative vector c , set $M = \max_{i,j} |(H_d)_{ij}|$ and $n = d + 1$. If $\|\tilde{c} - c\|_\infty \leq \varepsilon$, expansion gives

$$|\tilde{c}^T H_d \tilde{c} - c^T H_d c| \leq M(2\|c\|_1 n \varepsilon + n^2 \varepsilon^2). \quad (\text{WBR21})$$

If the original value is $-\mu < 0$, choose ε making the right side less than μ , then choose rational coefficients within that distance. This proves the rational-witness assertion with the original matrix fixed. The shifted family has quadratic form $\sum m_j \beta_j^2 P(\beta_j)^2$, and is also nonnegative under RH.

WBR5. Actual packet and its complete representative

Let Z be a nonempty finite set of actual nontrivial zeros of g , stable under reflection and conjugation, with every full order retained. Write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d_h = \deg h, \quad E_h = \mathbb{C}[s]/(h), \quad A_h = M_s, \quad \nu_h = g/h, \quad \eta_h = j_h(1/\nu_h). \quad (\text{WBR22})$$

The map j_h keeps all Taylor coefficients through orders $m_\rho - 1$. The element η_h is a unit because every order in h is the full order in g . Let $R_Z(u)$ be the unique polynomial of degree below d_h representing $\eta_h u$.

The original Euler inverse on a zero Mellin value is

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy, \quad (D - \rho)S_\rho F = F, \quad \mathcal{M}S_\rho F(s) = \frac{\mathcal{M}F(s)}{s - \rho}. \quad (\text{WBR23})$$

The zero Mellin value makes the upper integral the negative lower integral. At infinity the bound $F(x) \leq Cx^{-N}$ yields $C/(N - \Re \rho)$; at zero $F(x) \leq C'x^N$ yields $C'/(N + \Re \rho)$, when the denominators are positive. These inequalities, and $DS_\rho F = \rho S_\rho F + F$, prove every required seminorm bound. The homogeneous function $x^{-\rho}$ is excluded by the two-ended decay, so the inverse is unique and $D - \rho$ is injective. Iteration gives S_h on the vanishing full Mellin jets, independent of inversion order by uniqueness.

The actual section and observation are

$$\begin{aligned} R_{\text{ref}} u &= S_h \Theta(R_Z(u)(D)\phi_*), \quad \sigma_h u = [R_{\text{ref}} u] \in Q, \quad J_h F = j_h \mathcal{M}F, \\ \mathcal{M}R_{\text{ref}} u &= \nu_h R_Z(u), \quad J_h R_{\text{ref}} = 1_{E_h}, \quad J_h \Theta = 0. \end{aligned} \quad (\text{WBR24})$$

The transform $gR_Z(u)$ has all the zero jets needed by S_h . The section identity is $j_h \nu_h \eta_h = 1$. Thus σ_h is injective. Polynomial division gives

$$DR_{\text{ref}} - R_{\text{ref}} A_h = f_0 \ell_h, \quad \ell_h(u) = [s^{d_h-1}] R_Z(u). \quad (\text{WBR25})$$

Indeed $sR_Z(u) - R_Z(su) = h(s)\ell_h(u)$; the two sides of (WBR25) have equal Mellin transforms. Mellin injectivity follows from Fourier injectivity in logarithmic coordinates. Passing to the original quotient proves $D_Q \sigma_h = \sigma_h A_h$.

The source sequence cited by the package is

$$0 \longrightarrow Q[h(D)] \longrightarrow E_h \xrightarrow{\times j_h g} E_h \longrightarrow Q/h(D)Q \longrightarrow 0. \quad (\text{WBR26})$$

Its complete proof, including the source Euler inverse, full-jet surjectivity and the actual arrows, is included verbatim in dependencies/arithmetic_input.tex, (A5)–(A10). Equations (WBR23)–(WBR25) also prove directly every packet inclusion and operator identity used by this witness, with the full η_h .

WBR6. Reciprocal action, inverse jet, and literal theta boundary

Since $g(1/2) > 0$, $A_h - \frac{1}{2}I$ is invertible. Define on this finite packet

$$T_h = -(A_h - \frac{1}{2}I)^{-2}. \quad (\text{WBR27})$$

Through σ_h , this is also an endomorphism of the actual finite image in Q . The construction has this precise domain; no inverse on unrelated quotient classes is required.

At a primary block write $\lambda_\rho = \rho - 1/2$ and $A_h = \rho I + N_\rho$, with full nilpotent length m_ρ . The exact finite expansion is

$$T_h = \sum_{j=0}^{m_\rho-1} (-1)^{j+1} (j+1) \lambda_\rho^{-j-2} N_\rho^j. \quad (\text{WBR28})$$

This is the product $-\lambda_\rho^{-2}(1 + N_\rho/\lambda_\rho)^{-2}$. Multiplying the series by $(\lambda_\rho + N_\rho)^2$ gives constant -1 and zero at every positive degree below m_ρ , which proves the identity in the truncated algebra.

Put $\beta_\rho = -\lambda_\rho^{-2}$ and $Y_\rho = T_h - \beta_\rho I$. The coefficient of N_ρ is $2\lambda_\rho^{-3} \neq 0$. The complete inverse coordinate is

$$N_\rho = \lambda_\rho \sum_{j=1}^{m_\rho-1} \binom{-1/2}{j} (Y_\rho/\beta_\rho)^j. \quad (\text{WBR29})$$

This is a finite algebra identity without any analytic branch choice. In the truncated algebra the element U with constant term one and $U^2(1 + Y_\rho/\beta_\rho) = 1$ is unique: each successive coefficient has leading scalar 2. The displayed binomial coefficients solve these recursive equations. The element $1 + N_\rho/\lambda_\rho$ solves the same equation since $T_h/\beta_\rho = (1 + N_\rho/\lambda_\rho)^{-2}$, proving (WBR29). For $m_\rho = 1$, the positive-degree sums are empty and $N_\rho = 0$. Thus every original local length is retained.

There is a literal original-source representative of this action. Let $t_h(s)$ be the unique polynomial of degree below d_h representing $-(s - 1/2)^{-2}$ in E_h . For a polynomial P , let $p_h(s)$ be the degree-below- d_h representative of $P(t_h(s))$. Define

$$K_{P,h}(u; s) = \frac{p_h(s)R_Z(u)(s) - R_Z(P(T_h)u)(s)}{h(s)}. \quad (\text{WBR30})$$

The numerator vanishes modulo h because multiplication by η_h commutes with $P(T_h)$. Thus the quotient is a polynomial in the original variable s . Mellin transformation gives the exact original boundary

$$p_h(D)R_{\text{ref}}u - R_{\text{ref}}P(T_h)u = \Theta(K_{P,h}(u; D)\phi_*). \quad (\text{WBR31})$$

Both transforms are $gK_{P,h}(u; s)$, proving the formula and every sign. It specifies the difference of representatives before passing to Q , and proves the finite packet intertwining there. The support lift keeps the label of that theta boundary fixed.

WBR7. Full finite algebra, half-trace, and scalar radical

Let $\mathcal{B}_Z$ be the set of distinct reciprocal coordinates in this packet. A fibre of $\rho \mapsto \beta_\rho$ consists exactly of the distinct pair $\rho, 1 - \rho$: equality of the reciprocal squares gives equality or negation of $\rho - 1/2$. Their orders agree by reflection. Call this common order m_β , and put

$$f_Z(X) = \prod_{\beta \in \mathcal{B}_Z} (X - \beta)^{m_\beta}, \quad \mathcal{A}_Z = \mathbb{C}[X]/(f_Z). \quad (\text{WBR32})$$

The map $P \mapsto P(T_h)$ has kernel exactly (f_Z) . On each primary block, the inverse (WBR29) identifies the local algebra with $\mathbb{C}[Y]/Y^{m_\rho}$. Consequently the action vanishes precisely when every Taylor coefficient of P at β_ρ below that order is zero. Intersecting the coprime primary ideals proves the stated kernel and the embedding $\mathcal{A}_Z = \mathbb{C}[T_h] \subset \text{End}(E_h)$.

Choose one point of each reflected pair to write coordinates. Applying (WBR29) on both primary blocks and then Chinese remainders gives an explicit $\mathcal{A}_Z$ -module isomorphism

$$E_h \simeq \mathcal{A}_Z \oplus \mathcal{A}_Z. \quad (\text{WBR33})$$

The action of X on each copy is multiplication by X . Both reflected blocks remain in this display. Changing the coordinate choice swaps the corresponding two components without altering E_h or T_h .

On the local basis $1, Y, \dots, Y^{m_\beta-1}$, multiplication by $Q(\beta + Y)$ is triangular with constant diagonal $Q(\beta)$. Therefore

$$\begin{aligned} \frac{1}{2} \text{Tr}_{E_h} Q(T_h) &= \text{Tr}_{\mathcal{A}_Z} M_Q = \sum_{\beta \in \mathcal{B}_Z} m_\beta Q(\beta), \\ \frac{1}{2} \text{Tr}_{E_h} (T_h P(T_h)^2) &= \sum_{\beta \in \mathcal{B}_Z} m_\beta \beta P(\beta)^2. \end{aligned} \quad (\text{WBR34})$$

This proves the original factor $1/2$, the full multiplicity and the source-to-scalar map. Exhaustion by finite stable packets converges to (WBR14) by (WBR10).

The precise scalar information loss is also computable. On $\mathcal{A}_Z$, define the bilinear form

$$\mathbf{b}_Z(U, V) = \text{Tr}_{\mathcal{A}_Z} M_{XUV} = \sum_{\beta} m_\beta \beta U(\beta) V(\beta). \quad (\text{WBR35})$$

Its radical is exactly

$$\mathbf{n}_Z = \{U : U(\beta) = 0 \text{ for every } \beta\} = (\text{rad } f_Z)/(f_Z). \quad (\text{WBR36})$$

One inclusion follows from (WBR35). If $U(\beta_0) \neq 0$, choose a polynomial V equal to one at β_0 and zero at every other node, by Lagrange interpolation. Then the bilinear value is $m_{\beta_0} \beta_0 U(\beta_0) \neq 0$, proving the other inclusion. Thus the scalar form factors through the exact quotient $\mathcal{A}_Z \rightarrow \prod_{\beta} \mathbb{C}$, whose kernel is the displayed nilradical. Before that quotient, (WBR28)–(WBR33) retain every nilpotent coefficient; the trace weights retain every algebraic multiplicity. This specifies the full map relating the local algebra to its scalar observation.

WBR8. Exact critical-line comparison

The inherited source map is

$$\Gamma : Q \longrightarrow Q_{\text{crit}} = \mathcal{S}(\mathbb{R}) / \overline{\zeta(1/2 + it)\mathcal{S}(\mathbb{R})}, \quad \Gamma[F] = [\mathcal{M}F(1/2 + it)]. \quad (\text{WBR37})$$

Its construction and range are included whole in dependencies/HOCHSCHILD_COMPARISON.md, (H16)–(H25). For $\Re \rho \neq 1/2$, multiplication by $(1/2 + it - \rho)^{-1}$ is a continuous Schwartz multiplier, preserves the displayed relation and its closure, and inverts $M_{1/2+it} - \rho$ on Q_{crit} . Intertwining by D therefore sends every full generalized ρ -block to zero under $\Gamma\sigma_h$.

At a critical zero $\rho = 1/2 + i\gamma$ of full order m , the continuous jet map

$$[f] \mapsto \sum_{j=0}^{m-1} \frac{i^{-j} f^{(j)}(\gamma)}{j!} \epsilon^j \quad (\text{WBR38})$$

vanishes on the relation and its closure. On $\Gamma[R_{\text{ref}}u]$, it equals $j_\rho u$: each t -derivative gives the factor i^j in the Mellin derivative. Together these jet maps invert every critical block. Thus

$$\ker(\Gamma\sigma_h) = E_{h,\text{off}}. \quad (\text{WBR39})$$

In the reciprocal algebra, the corresponding actual map is the Chinese-remainder projection

$$\pi_{\text{crit}} : \mathcal{A}_Z \longrightarrow \mathcal{A}_{Z,\text{crit}}, \quad \ker \pi_{\text{crit}} = \mathcal{A}_{Z,\text{off}}. \quad (\text{WBR40})$$

The functional (WBR34) decomposes into its critical and off-critical summands. If an off-critical node β occurs, its primary idempotent e_β , equal to one on its entire local block and zero on the others, belongs to $\ker \pi_{\text{crit}}$, yet has $\text{Tr}_{\mathcal{A}_Z} M_{X_{e_\beta}} = m_\beta \beta \neq 0$. Hence in that case the full functional does not factor through π_{crit} . Its exact omitted part is the displayed weighted trace on $\mathcal{A}_{Z,\text{off}}$. On real polynomials the two-value interpolation (WBR16) selects the conjugate pair and kills every other finite packet node, giving the value $-2m$; the actual infinite arithmetic tail is controlled by (WBR18).

All these maps lift with the source label fixed. Thus an off-critical vector maps to $(\ell, 0)$ under the split lift of $\Gamma\sigma_h$, while remaining the input (ℓ, v) of the full source operator and its trace. The exact maps (WBR29)–(WBR40) relate the arithmetic source, the scalar witness and the receiving quotient.

WBR9. Computation and integration record

The original integral in the numerical script is

$$M_{2j} = 4 \sum_{n \geq 1} \int_0^\infty y^{2j} (4\pi^2 n^4 e^{9y/2} - 6\pi n^2 e^{5y/2}) e^{-\pi n^2 e^{2y}} dy. \quad (\text{WBR41})$$

The factor four is the two in Θ and the two half-lines under reflection. The supplied script retains these constants. It uses finite theta sums and the interval $0 \leq y \leq 4$. Its two high-precision runs have no validated enclosure of the finite quadrature or omitted tails.

The finite independent replay verifies the saved decimal matrices by exact rational LDL arithmetic. Those matrices are positive through dimension 16 in both shifts. This is a theorem about the specified decimal rational matrices. The map from those matrices to actual H_d requires coefficient enclosures; none is supplied by agreement between two floating runs.

The supplied verifier also permits negative shift values, using Python negative indices, and the coefficient recurrence uses floating division when passed integer inputs. These narrow input-contract defects do not affect the archived default runs or the mathematical recurrence. The additive strict verifier accepts explicit exact rationals, validates nonnegative integral indices and interval order, and leaves the archive unchanged. The full replay report, exact tests and machine receipts accompany this proof.

The source NOTE equations (1)–(23), with the explicit $\nu_N = u_N$ symbol repair above, are accepted with their stated standard Hadamard input. For cumulative integration, include its entire original LaTeX, this entire proof, the complete finite replay and both inherited dependency files. Place the reciprocal action after the original arithmetic packet, and its Hankel scalar after the original theta moments. Its source class remains available to the ongoing metric and absolute-base geometric calculations.

Classical context was checked directly at Zhang’s author record. This lies in the established power-sum approach. The current proof makes no novelty claim for that general approach and does not depend on an uninspected theorem from that paper.

Certified original-theta Hankel calculation

Certified original-theta Hankel calculation

This calculation certifies strict positive definiteness of both original matrices

$$\mathbf{H}_{15} = (b_{r+s})_{0 \leq r, s \leq 15}, \quad \mathbf{H}_{15}^{(1)} = (b_{r+s+1})_{0 \leq r, s \leq 15}. \quad (\text{CH.1})$$

Their coefficients are computed from the original $g = 2\xi$ theta source. Every one of their 32 leading principal minors is certified positive. The coordinates, original mass, theta terms and logarithmic coefficient recurrence are retained. No zero ordinate is supplied to the calculation.

1. Original integral and coefficients

Use the original functions

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad f_0(x) = 2 \sum_{n \geq 1} \phi_*(nx), \quad \psi(y) = e^{y/2} f_0(e^y). \quad (\text{CH.2})$$

For completeness, write $\vartheta(x) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x^2}$ and $D = -x \partial_x$. Termwise differentiation on compact positive x -intervals gives $f_0 = (D^2 - D)\vartheta$. The original Gaussian Poisson identity is $\vartheta(x) = x^{-1} \vartheta(1/x)$. For the exact reflection operator $(\mathcal{R}v)(x) = x^{-1} v(1/x)$, differentiation gives $D\mathcal{R} = \mathcal{R}(1-D)$, and hence $(D^2 - D)\mathcal{R} = \mathcal{R}(D^2 - D)$. Applying this identity to ϑ proves $f_0(x) = x^{-1} f_0(1/x)$ and therefore $\psi(-y) = \psi(y)$. All powers of x are retained in this map. Consequently, for every even nonnegative integer J ,

$$\begin{aligned} M_J &= \int_{\mathbb{R}} y^J \psi(y) dy \\ &= \int_0^\infty y^J \sum_{n \geq 1} (16\pi^2 n^4 e^{9y/2} - 24\pi n^2 e^{5y/2}) e^{-\pi n^2 e^{2y}} dy. \end{aligned} \quad (\text{CH.3})$$

Both factors two, from theta summation and reflection, are present. Every summand in (CH.3) is nonnegative for $y \geq 0$, and is strictly positive for $y > 0$: its non-polynomial coefficient factors as

$$8\pi n^2 e^{5y/2} (2\pi n^2 e^{2y} - 3) e^{-\pi n^2 e^{2y}} > 0. \quad (\text{CH.4})$$

For $J = 0$, positivity holds at $y = 0$ as well.

The Mellin substitution $x = e^y$, including $dx = e^y dy$, gives

$$g(1/2 + iz) = \int_{\mathbb{R}} \psi(y) e^{izy} dy, \quad a_j = \frac{(-1)^j M_{2j}}{(2j)!}, \quad F(w) = \sum_{j \geq 0} a_j w^j. \quad (\text{CH.5})$$

The bounds proved below supply absolute integrability after multiplication by every power of y , and by $e^{A|y|}$ for each fixed A : replace $9/2$ by $9/2 + A$ in (CH.11) and choose a sufficiently large positive L , so its decay constant is positive. Reflection gives the same estimate on the other half-line. Thus dominated differentiation on compact z -sets is valid and proves $F(z^2) = g(1/2 + iz)$ with the displayed coefficients. In particular the original $a_0 = M_0 = g(1/2)$ is strictly positive. Multiplication of the convergent power series $-F'(w)/F(w) = \sum_{n \geq 0} b_n w^n$ by $F(w)$ gives

$$a_0 b_n + \sum_{j=1}^n a_j b_{n-j} = -(n+1) a_{n+1}, \quad b_n = \frac{-(n+1) a_{n+1} - \sum_{j=1}^n a_j b_{n-j}}{a_0}. \quad (\text{CH.6})$$

This is the exact coefficient map used by the program. The denominator is the computed interval for the original a_0 . It is never replaced by one.

2. Explicit positive omitted regions

Fix $N = 20$ and $L = 4$. The finite integral retains precisely $1 \leq n \leq 19$ and $0 \leq y \leq 4$. Its complement is the disjoint union of the omitted terms $n \geq 20$ on $0 \leq y \leq 4$ and every $n \geq 1$ on $y > 4$. Endpoints have measure zero.

For $n = N + \ell$, $\ell \geq 0$, the elementary inequality $\log(1 + t) \leq t$ for $t \geq 0$, proved by differentiating $t - \log(1 + t)$, implies

$$(N + \ell)^4 \leq N^4 e^{4\ell/N}, \quad (N + \ell)^2 \geq N^2 + 2N\ell. \quad (\text{CH.7})$$

Hence, for $y \geq 0$,

$$\sum_{n \geq N} n^4 e^{-\pi n^2 e^{2y}} \leq N^4 e^{-\pi N^2} \sum_{\ell \geq 0} e^{(4/N - 2\pi N)\ell} = \frac{N^4 e^{-\pi N^2}}{1 - e^{4/N - 2\pi N}}. \quad (\text{CH.8})$$

The exponent $4/N - 2\pi N$ is negative for every integer $N \geq 1$. Using (CH.4), and bounding the positive difference in (CH.3) by its first positive term, the first omitted region lies in $[0, T_{J,n}]$, where

$$T_{J,n} = \frac{16\pi^2 L^{J+1} e^{9L/2} N^4 e^{-\pi N^2}}{1 - e^{4/N - 2\pi N}}. \quad (\text{CH.9})$$

Indeed $y^J \leq L^J$ and $e^{9y/2} \leq e^{9L/2}$ on the finite interval, whose length contributes the displayed factor L . The deliberately larger bound L^{J+1} is retained.

For the second omitted region the same argument, with $N = 1$ but retaining the y -dependent first exponential, gives

$$\sum_{n \geq 1} n^4 e^{-\pi n^2 e^{2y}} \leq \frac{e^{-\pi e^{2y}}}{1 - e^{4 - 2\pi}}, \quad y \geq 0. \quad (\text{CH.10})$$

Set $h_J(y) = y^J e^{9y/2 - \pi e^{2y}}$ on $y \geq L > 0$. Its logarithmic derivative is

$$\frac{h'_J(y)}{h_J(y)} = \frac{J}{y} + \frac{9}{2} - 2\pi e^{2y} \leq -\alpha_J, \quad \alpha_J = 2\pi e^{2L} - \frac{9}{2} - \frac{J}{L}. \quad (\text{CH.11})$$

For $0 \leq J \leq 64$, $N = 20$, $L = 4$, all α_J are strictly positive. For example, $\pi > 3$ and $e^8 > 1 + 8 + 8^2/2 = 41$ give $\alpha_J > 246 - 9/2 - 16 > 0$. Integration of (CH.11) gives $h_J(y) \leq h_J(L) e^{-\alpha_J(y-L)}$, and then

$$T_{J,y} = \frac{16\pi^2 L^J e^{9L/2 - \pi e^{2L}}}{(1 - e^{4 - 2\pi})(2\pi e^{2L} - 9/2 - J/L)} \quad (\text{CH.12})$$

bounds the second omitted region. Therefore the complete exact moment satisfies

$$M_J \in I_{J,\text{finite}} + [0, T_{J,n} + T_{J,y}]. \quad (\text{CH.13})$$

The constants and signs of both original terms survive (CH.3) and the finite integration. The positive upper majorant is used only for the two explicitly omitted regions.

3. Validated finite integration and its exact export

The program uses python-flint 0.9.0, linked to FLINT 3.6.0, with 1024-bit arithmetic. For each even $J = 0, 2, \dots, 64$, it evaluates

$$z^J e^{z/2} \sum_{n=1}^{19} (16(\pi n^2 e^{2z})^2 - 24\pi n^2 e^{2z}) e^{-\pi n^2 e^{2z}}. \quad (\text{CH.14})$$

This is algebraically identical to the finite integrand in (CH.3). It is an entire complex function: all powers are nonnegative integer powers, and every remaining operation is addition, multiplication or the entire exponential. It has no branch cut, pole, absolute value, conjugation or piecewise real continuation.

The `acb.integral` callback receives a Boolean `analytic`. The official contract requires the callback to establish analyticity on the input complex ball when this flag is true. Formula (CH.14) is entire, so the same formula is valid for both values of the flag on every ball. This is the reason that the implementation may ignore the flag. It is not assuming that analyticity along the real segment alone suffices.

The finite interval is divided at the exact rationals $0, 1/2, 1, \dots, 7/2, 4$. Each of its eight integrals is evaluated with relative and absolute tolerance goals 2^{-850} , evaluation limit 200000 and depth limit 50. These controls affect effort and enclosure width; the final proof uses the returned enclosures themselves. Every returned value must be finite and its imaginary part must contain zero. Its real interval therefore encloses the real integral. The eight real intervals are added with outward rounding.

The two tail formulas are evaluated with Arb arithmetic, after verifying strict positivity of each denominator interval. If their sum has upper endpoint U_J , the program forms the interval hull $(0, U_J)$, and adds it to the real finite interval. This proves (CH.13) with the actually stored enclosure.

For every reported Arb ball, `lower()` and `upper()` produce exact dyadic endpoints rounded respectively downward and upward. `man_exp()` writes each as $m2^e$, with integers m, e . The export writes that same rational as numerator $m2^e$ and denominator one if $e \geq 0$, and numerator m , denominator 2^{-e} otherwise. No decimal conversion intervenes. Every decimal display field is for reading only. The lower and upper integer fractions are the authoritative certificate. The program raises the Python integer-string limit to accommodate the large dyadic denominator of (CH.12); this changes only serialization capacity.

4. Interval coefficient and LDL proof

Arb interval operations enclose every real operation on represented inputs. Induction in (CH.6), beginning with $a_0 > 0$, therefore proves that each stored a_j encloses the exact coefficient in (CH.5) and each stored b_n encloses the exact logarithmic coefficient in (CH.6). The 33 moments determine $a_0, \dots, a_{32}$ and $b_0, \dots, b_{31}$, exactly the data required by (CH.1).

For each shift $\epsilon = 0, 1$, put $A_{ij} = b_{i+j+\epsilon}$. In the original ordered monomial basis, define the unit lower triangular matrix L and diagonal $D = \text{diag}(d_0, \dots, d_{15})$ recursively by

$$d_j = A_{jj} - \sum_{k < j} L_{jk}^2 d_k, \quad L_{ij} = \frac{A_{ij} - \sum_{k < j} L_{ik} L_{jk} d_k}{d_j} \quad (i > j). \quad (\text{CH.15})$$

The implementation checks that the entire computed interval for d_j is strictly positive before every division. Induction on j proves that the stored interval values contain the exact values of (CH.15). The equations for the diagonal and lower triangular entries are exactly the equations of $A = LDL^T$; symmetry supplies the upper triangular entries. Since L is invertible with determinant one,

$$c^T A c = \sum_{j=0}^{15} d_j (L^T c)_j^2 > 0 \quad \text{for every } c \in \mathbb{R}^{16} \setminus \{0\}. \quad (\text{CH.16})$$

For each leading size r , restriction to the first r coordinates and triangularity give

$$\det A_{[0, r-1]} = \prod_{j=0}^{r-1} d_j > 0. \quad (\text{CH.17})$$

The exact interval computation certifies all 32 pivots as positive. No pivoting, coordinate rescaling, diagonal congruence or replacement of the source mass occurs.

5. Recorded values and finite conclusion

The authoritative file `certificate_dimension16_1024.json` stores all 264 finite integral pieces, their real and imaginary intervals, all 66 tail bounds, all 33 complete moments, all coefficients, and both complete lower triangular factor intervals. The run took about 16.14 seconds on the recorded host. This duration measures one reproducible computation, rather than a mathematical assumption.

For orientation the original mass and first coefficients begin

$$\begin{aligned} a_0 &= 0.99424155637662821982554747937079544 \dots, \\ b_0 &= 0.023104993115418970788933810430339014 \dots, \\ b_1 &= 3.7172599285269686164866262471740578 \dots \times 10^{-5}, \\ b_2 &= 1.4417393140097327969538155609482091 \dots \times 10^{-7}. \end{aligned} \quad (\text{CH.18})$$

The last pivots and full determinants are enclosed within the following simple rational intervals:

$$\begin{array}{c|c|c} \epsilon & d_{15} & \det H_{15}^{(\epsilon)} \\ \hline 0 & (6 \cdot 10^{-107}, 7 \cdot 10^{-107}) & (9 \cdot 10^{-822}, 10 \cdot 10^{-822}) \\ 1 & (7 \cdot 10^{-111}, 8 \cdot 10^{-111}) & (6 \cdot 10^{-878}, 7 \cdot 10^{-878}) \end{array} \quad (\text{CH.19})$$

The exact stored rational endpoints are much narrower; the respective last pivots retain 656 and 648 relative accuracy bits.

Thus this calculation establishes strict positivity of the two specified dimension-16 matrices and every leading truncation. Its quantified conclusion concerns these finite matrices. It supplies certified original-theta inputs in place of the Workbench's unvalidated quadrature values at these dimensions.

Reproduction

Run `certify_theta_hankel.py` with arguments `--dimension 16 --precision 1024 --target-bits 850 --output certificate_dimension16_1024.json` using `python-flint 0.9.0`. The defaults retain $N = 20, L = 4$. The file `certificate_dimension2_256.json` records a separate smaller run with dimension two, precision 256 and tolerance exponent 200.

The official integration and real-endpoint API contracts are documented at:

- `python-flint` complex integration
- `python-flint` real ball endpoints

The local `API_CONTRACT.txt` records the installed method docstrings used when implementing the computation.

Independent theta-tail and certificate-code proof

Independent theta-tail proof and certificate code review

Date: 2026-09-13.

This review concerns the exact original theta integrand, its omitted tails, and the complete code path that produces the dimension-16 logarithmic Hankel certificate. It retains every coefficient in the integrand. The parent is independently replaying the rational interval arithmetic; this review supplies the analytic proof and an independent complete source read.

1. Exact objects and retained integral

Let J be a nonnegative integer, N a positive integer, and $L > 0$. For $n \geq 1$ and real $y \geq 0$, set

$$f_{J,n}(y) = y^J (16\pi^2 n^4 e^{9y/2} - 24\pi n^2 e^{5y/2}) e^{-\pi n^2 e^{2y}}, \quad F_J(y) = \sum_{n=1}^{\infty} f_{J,n}(y). \quad (\text{TR1})$$

When $J = 0$, the factor y^0 is the constant function 1, including at zero. The intended integral and retained part are

$$I_J = \int_0^{\infty} F_J(y) dy, \quad I_{J,N,L}^{\text{fin}} = \int_0^L \sum_{n=1}^{N-1} f_{J,n}(y) dy. \quad (\text{TR2})$$

For $N = 1$, the finite sum is empty and its integral is zero. The actual run uses $N = 20$, $L = 4$, and $J = 0, 2, \dots, 64$; it therefore retains precisely $n = 1, \dots, 19$.

2. Positivity and the exact majorant

The identity

$$16\pi^2 n^4 e^{9y/2} - 24\pi n^2 e^{5y/2} = 8\pi n^2 e^{5y/2} (2\pi n^2 e^{2y} - 3) \quad (\text{TR3})$$

proves that the parenthesis in (TR1) is strictly positive, since $2\pi n^2 e^{2y} - 3 \geq 2\pi - 3 > 0$. Thus

$$0 \leq f_{J,n}(y) \leq 16\pi^2 y^J e^{9y/2} n^4 e^{-\pi n^2 e^{2y}}. \quad (\text{TR4})$$

The upper estimate discards a nonnegative subtracted term only in the estimate. Both terms remain in the actual finite integral.

For every $N \geq 1$,

$$0 < r_N := e^{4/N - 2\pi N} \leq r_* := e^{4 - 2\pi} < 1. \quad (\text{TR5})$$

Here $4/N \leq 4$, $N \geq 1$, and $\pi > 3$.

3. Omitted summands on the retained interval

Put $a_n = n^4 e^{-\pi n^2}$. Its exact consecutive-term ratio is

$$\frac{a_{n+1}}{a_n} = \left(1 + \frac{1}{n}\right)^4 e^{-\pi(2n+1)}. \quad (\text{TR6})$$

For $x \geq 0$, $\log(1+x) \leq x$: the difference $x - \log(1+x)$ is zero at zero and has derivative $x/(1+x) \geq 0$. Consequently, for $n \geq N$,

$$\begin{aligned} \log \frac{a_{n+1}}{a_n} &= 4 \log(1 + 1/n) - 2\pi n - \pi \\ &\leq 4/n - 2\pi n - \pi \\ &\leq 4/N - 2\pi N - \pi \leq 4/N - 2\pi N = \log r_N. \end{aligned} \quad (\text{TR7})$$

It follows by induction that $a_{N+j} \leq a_N r_N^j$ for every integer $j \geq 0$, and hence

$$\sum_{n=N}^{\infty} n^4 e^{-\pi n^2} \leq \frac{N^4 e^{-\pi N^2}}{1 - r_N}. \quad (\text{TR8})$$

On $0 \leq y \leq L$, use $y^J \leq L^J$, $e^{9y/2} \leq e^{9L/2}$, and $e^{-\pi n^2 e^{2y}} \leq e^{-\pi n^2}$. Equations (TR4) and (TR8) then give

$$\begin{aligned} 0 \leq E_{J,N,L}^{(n)} &:= \int_0^L \sum_{n=N}^{\infty} f_{J,n}(y) dy \\ &\leq \frac{16\pi^2 L^{J+1} e^{9L/2} N^4 e^{-\pi N^2}}{1 - e^{4/N-2\pi N}} =: T_n(J, N, L). \end{aligned} \quad (\text{TR9})$$

The extra negative term $-\pi$ in (TR7) was discarded in the direction that enlarges the upper bound. This proves the proposed expression with its stated denominator.

The same argument with $N = 1$ gives a summable uniform majorant on each compact interval $[0, L]$. The series defining F_J is therefore uniformly convergent there and continuous. Its summands are nonnegative, so monotone convergence also justifies the sum-integral interchanges above and below.

4. The complete integration tail

For fixed $y \geq 0$, write $a_n(y) = n^4 e^{-\pi n^2 e^{2y}}$. Then

$$\begin{aligned} \log \frac{a_{n+1}(y)}{a_n(y)} &= 4 \log(1 + 1/n) - \pi(2n+1)e^{2y} \\ &\leq 4/n - 2\pi n e^{2y} - \pi e^{2y} \\ &\leq 4 - 2\pi = \log r_*. \end{aligned} \quad (\text{TR10})$$

The last inequality uses $1/n \leq 1$, $n e^{2y} \geq 1$, and $-\pi e^{2y} \leq 0$. Since $a_1(y) = e^{-\pi e^{2y}}$, the geometric-series estimate is

$$\sum_{n=1}^{\infty} n^4 e^{-\pi n^2 e^{2y}} \leq \frac{e^{-\pi e^{2y}}}{1 - e^{4-2\pi}}. \quad (\text{TR11})$$

Therefore

$$0 \leq F_J(y) \leq \frac{16\pi^2}{1 - e^{4-2\pi}} y^J e^{9y/2 - \pi e^{2y}}. \quad (\text{TR12})$$

Define

$$A_{J,L} = 2\pi e^{2L} - \frac{9}{2} - \frac{J}{L}. \quad (\text{TR13})$$

When $A_{J,L} > 0$, the positive function $q_J(y) = y^J e^{9y/2 - \pi e^{2y}}$, for $y \geq L > 0$, satisfies

$$\frac{q'_J(y)}{q_J(y)} = \frac{J}{y} + \frac{9}{2} - 2\pi e^{2y} \leq \frac{J}{L} + \frac{9}{2} - 2\pi e^{2L} = -A_{J,L}. \quad (\text{TR14})$$

The first summand is identically zero for $J = 0$. Integrating this differential inequality gives $q_J(y) \leq q_J(L) e^{-A_{J,L}(y-L)}$. Thus

$$\begin{aligned} 0 \leq E_{J,L}^{(y)} &:= \int_L^{\infty} \sum_{n=1}^{\infty} f_{J,n}(y) dy \\ &\leq \frac{16\pi^2 L^J e^{9L/2 - \pi e^{2L}}}{(1 - e^{4-2\pi})(2\pi e^{2L} - 9/2 - J/L)} =: T_y(J, L). \end{aligned} \quad (\text{TR15})$$

This integral includes every $n \geq 1$, including all omitted summands $n \geq N$ beyond L .

5. The exact remainder and interval addition

The omitted domain is the disjoint union, up to the measure-zero endpoint slice, of

$$\{(n, y) : n \geq N, 0 \leq y < L\} \quad \text{and} \quad \{(n, y) : n \geq 1, L \leq y < \infty\}. \quad (\text{TR16})$$

The two integrals are nonnegative and finite by (TR9) and (TR15). They therefore give the exact identity and enclosure

$$I_J = I_{J,N,L}^{\text{fin}} + E_{J,N,L}^{(n)} + E_{J,L}^{(y)}, \quad I_J - I_{J,N,L}^{\text{fin}} \in [0, T_n + T_y]. \quad (\text{TR17})$$

There is no double counting of $n \geq N$ beyond L : that region occurs only in $E^{(y)}$. If $[a, b]$ encloses the finite integral and $\bar{T} \geq T_n + T_y$, then

$$I_J \in [a, b] + [0, \bar{T}] = [a, b + \bar{T}]. \quad (\text{TR18})$$

The entire interval $[0, \overline{T}]$, rather than a tail midpoint, is what must be added.

For the actual parameters the formulas are

$$\begin{aligned} T_n(J, 20, 4) &= \frac{16\pi^2 4^{J+1} e^{18} 20^4 e^{-400\pi}}{1 - e^{1/5-40\pi}}, \\ T_y(J, 4) &= \frac{16\pi^2 4^J e^{18-\pi e^8}}{(1 - e^{4-2\pi})(2\pi e^8 - 9/2 - J/4)}. \end{aligned} \quad (\text{TR19})$$

All denominator signs are proved exactly for $0 \leq J \leq 64$:

$$1/5 - 40\pi < -599/5 < 0, \quad 4 - 2\pi < -2 < 0, \quad (\text{TR20})$$

and $e^8 > 1 + 8 = 9$ gives

$$2\pi e^8 - \frac{9}{2} - \frac{J}{4} > 6 \cdot 9 - \frac{9}{2} - 16 = \frac{67}{2} > 0. \quad (\text{TR21})$$

Thus the claimed tail hypotheses are verified for every actual moment in the run, without a floating-point sign assumption.

6. Entire finite integrand and its literal code expression

For each nonnegative integer J , the function

$$z \mapsto z^J \sum_{n=1}^{N-1} (16\pi^2 n^4 e^{9z/2} - 24\pi n^2 e^{5z/2}) e^{-\pi n^2 e^{2z}} \quad (\text{TR22})$$

is entire on $\mathbb{C}$: the only operations are a polynomial, exponentials and their compositions, and finite sums and products. The code sets $x = \pi n^2 e^{2z}$ and evaluates

$$z^J e^{z/2} \sum_{n=1}^{N-1} (16x^2 - 24x) e^{-x}. \quad (\text{TR23})$$

The exact identities

$$16x^2 e^{z/2} = 16\pi^2 n^4 e^{9z/2}, \quad 24x e^{z/2} = 24\pi n^2 e^{5z/2} \quad (\text{TR24})$$

prove equality of (TR23) and (TR22). This rewrite retains the subtracted term and all original coefficients.

The installed python-flint 0.9.0 documentation for `acb.integral` was read directly. Its callback receives a complex ball and an analytic flag. When that flag is true, the callback must establish analyticity on the ball or return a nonfinite ball. The entire function (TR22) meets this requirement on every complex ball, so ignoring that flag in this particular callback is justified by the proof above. There are no fractional powers, logarithms, branches, or pole-containing denominators in the callback.

The code uses `arb.pi()` and integer arithmetic for its coefficients. Every division appearing in the callback or integration endpoints is performed on an `Arb/Acb` object and an exact integer. The finite path is partitioned into the eight consecutive exact intervals $[j/2, (j+1)/2]$, $j = 0, \dots, 7$. Their union is exactly $[0, 4]$. The code rejects nonfinite integrals and any reported imaginary enclosure excluding zero, then sums the complex balls. Since the exact integral is real, its real-part enclosure contains (TR2). The tolerance is an exact dyadic 2^{-850} ; the working precision is 1024 bits. The elapsed-time floats do not enter the mathematics.

7. Complete tail-code and exact-export review

The function `positive_tails` computes precisely (TR9), (TR15), and (TR13), with outward-rounded `Arb` operations. It explicitly rejects any run that fails to certify $1 - r_N > 0$, $1 - r_* > 0$, or $A_{J,L} > 0$. It uses the original factor $16\pi^2$ in each tail.

The function `compute_moment` takes the outward upper endpoint of the sum of the two tail balls, forms the ball union of that endpoint with zero, and adds that union to the real finite-integral ball. An enclosing ball union can slightly enlarge the endpoint interval, but it contains the whole interval required by (TR18). It does not replace the nonnegative tail interval with its midpoint. The final strict-positive comparison certifies positivity of the whole resulting moment ball.

The installed `arb.lower` and `arb.upper` documentation was also read directly. These methods return exact finite floating-point numbers rounded respectively toward negative and positive infinity at the active precision. The

installed `arb.man_exp` documentation says it requires an exact finite input and returns an integer mantissa m and integer exponent e . Therefore the exported endpoint is exactly

$$m2^e = \begin{cases} (m2^e)/1, & e \geq 0, \\ m/(2^{-e}), & e < 0. \end{cases} \quad (\text{TR25})$$

This is precisely the case distinction in `exact_dyadic`. Python integers and their decimal strings retain these numerators and denominators exactly. They need not be reduced to lowest terms to represent the same rational endpoint. The enclosing real interval exported by `enclosure` consequently retains a certified lower and upper rational bound; its shorter decimal display is additional presentation data.

The helper from `_enclosure` parses both endpoint fractions through `Arb` and unions their enclosures. This remains sound if rounding enlarges them. It is unused in the original integration and LDL run and therefore does not participate in the output's arithmetic chain.

8. Original logarithmic coefficient recurrence

Write $M_{2j} = I_{2j}$ for the original moments defined by (TR2). Define the exact formal series

$$A(z) = \sum_{j \geq 0} a_j z^j, \quad a_j = \frac{(-1)^j M_{2j}}{(2j)!}, \quad B(z) = -\frac{A'(z)}{A(z)} = \sum_{n \geq 0} b_n z^n. \quad (\text{TR26})$$

The nonzero constant term $a_0 = I_0 > 0$ is proved by (TR1), (TR3), and integration over any interval of positive length. Thus the formal division in (TR26) is uniquely defined, and no convergence assumption is needed to compute its finite jet. The identity $A(z)B(z) = -A'(z)$, in degree n , is

$$a_0 b_n + \sum_{j=1}^n a_j b_{n-j} = -(n+1)a_{n+1}. \quad (\text{TR27})$$

Solving for b_n gives exactly the recurrence in `logarithmic_coefficients`:

$$b_n = \frac{-(n+1)a_{n+1} - \sum_{j=1}^n a_j b_{n-j}}{a_0}. \quad (\text{TR28})$$

The factorials and signs are Python integers; the divisions and sums occur in `Arb` arithmetic. Starting from enclosing moment balls, induction on n proves that the computed a_j and b_n balls contain the exact coefficients (TR26). The denominator ball is certified positive before division. Correlations between occurrences of earlier balls can enlarge the resulting intervals but cannot invalidate their containment.

For dimension $d = 16$, the loop integrates $J = 0, 2, \dots, 4d$, giving $2d + 1 = 33$ moments and $a_0, \dots, a_{32}$. The recurrence produces $b_0, \dots, b_{31}$. Both Hankel matrices

$$H^{(r)} = (b_{i+j+r})_{0 \leq i, j < 16}, \quad r \in \{0, 1\}, \quad (\text{TR29})$$

are therefore completely specified: their largest needed indices are 30 and 31. The code uses these exact indices. It introduces no rescaling of the coefficient sequence or matrix.

9. Unpivoted LDL proof and the meaning of positivity

Fix $r \in \{0, 1\}$. For the exact real symmetric matrix $H^{(r)}$, let L be lower triangular with diagonal entries 1 and define, in increasing order of j ,

$$\begin{aligned} d_j &= H_{jj}^{(r)} - \sum_{k=0}^{j-1} L_{jk}^2 d_k, \\ L_{ij} &= \frac{H_{ij}^{(r)} - \sum_{k=0}^{j-1} L_{ik} L_{jk} d_k}{d_j}, \quad i > j. \end{aligned} \quad (\text{TR30})$$

The code implements these exact ordered recurrences. At step j , every lower entry used in the sums was computed at an earlier step. Induction on j proves that the stored balls enclose the exact values: the initial coefficient balls enclose $H^{(r)}$; sums and products preserve enclosure; and every divisor interval is certified strictly positive before its use. Hence each exact $d_j > 0$.

To establish the matrix identity, for $i = j$, the first recurrence in (TR30) gives $(LDL^\top)_{jj} = H_{jj}^{(r)}$; for $i > j$, the second gives $(LDL^\top)_{ij} = H_{ij}^{(r)}$. Symmetry supplies the entries $i < j$. Thus

$$H^{(r)} = L \operatorname{diag}(d_0, \dots, d_{15}) L^\top. \quad (\text{TR31})$$

Since L is unit lower triangular it is invertible. For every nonzero $v \in \mathbb{R}^{16}$, the vector $L^\top v$ is nonzero, and

$$v^\top H^{(r)} v = \sum_{j=0}^{15} d_j (L^\top v)_j^2 > 0. \quad (\text{TR32})$$

This proves positive definiteness of each original unscaled matrix, once all 16 pivot intervals are positive. Restricting (TR31) to the first s indices gives the exact leading determinant

$$\det H_{[0,s)}^{(r)} = \prod_{j=0}^{s-1} d_j, \quad 1 \leq s \leq 16, \quad (\text{TR33})$$

which is the interval product saved in each row. The unsuccessful branch reports `positivity_not_certified` rather than asserting that the matrix is indefinite. A finite useful certificate is produced only after all divisions used in it have positive divisor intervals.

10. Reviewed source and observed certificate

The complete 163-line source was read, including argument validation, partial-file writes, numerical functions, serialization, final status, and both matrix calls. The file reviewed is `certify_theta_hankel.py`, SHA-256:

a97486edc2fc1f05f155a9539ad280433be39a8f6924efbc8c1c1c4d71033950

The inspected certificate is `certificate_dimension16_1024.json`, SHA-256:

869ebacd3f82e46c3fe5fac1f07bcbaae7f4c333e1a5ba2e0ffcb1f91884a0fc

Its `source_sha256` equals the independently hashed reviewed source. It records python-flint 0.9.0, 1024-bit working precision, 850 target bits, dimension 16, $N = 20$, and $L = 4$. It contains all 33 required moments, eight finite-integration pieces per moment, 33 coefficient a intervals, 32 coefficient b intervals, and 16 successful pivot rows for each $r = 0, 1$. Its final status is `positive_definite_certified`.

The certificate's displayed dimension-16 endpoints, quoted here only as orientation while exact dyadic endpoints remain authoritative, are approximately:

Matrix	Last pivot	Full determinant
$H^{(0)}$	$6.1698420811114408502314906319786538 \times 10^{-107}$	$9.2735841166129968999872729909592731 \times 10^{-822}$
$H^{(1)}$	$7.1026100619440799659087832255229554 \times 10^{-111}$	$6.3606810193647631787476650057683745 \times 10^{-878}$

11. Acceptance and scope of the completed proof

Both proposed tail formulas are valid with every original constant. The analytic callback meets its contract, the retained summand matches the original expression exactly, the code adds the required nonnegative tail interval, the endpoint export is exact rational, the logarithmic recurrence has the correct signs and indices, and the LDL recurrences certify the original two matrices. This review found no formula/code mismatch and no correction necessary for the inspected dimension-16 run.

The exact objects certified are the moments (TR2), the finite logarithmic jet (TR26), and the two finite matrices (TR29). Their positive definiteness gives every leading determinant through dimension 16 by (TR33). Identifying a separately defined zeta function with A , or passing from these finite matrices to an assertion covering every dimension, requires the corresponding exact map and argument in the surrounding research; neither further statement is silently inserted into this certificate. The completed tail and finite-matrix calculations themselves use no zero ordinates, off-line zero assumption, or unproved infinite-dimensional positivity assertion.

Independent verification of the two positive theta tails

Independent verification of the positive theta tails

Date: 2026-09-13. The complete file `theta_certified_hankel_20260913/certify_theta_hankel.py` was read. Its SHA-256 is `a97486edc2fc1f05f155a9539ad280433be39a8f6924efbc8c1c1c4d71033950`. This review verifies the two formulas in `positive_tails` and the finite integration callback's analyticity. It does not rerun integration or review the separately assigned logarithmic recurrence, dyadic export, and LDL calculation.

Finding: both tail bounds are valid upper bounds with the printed constants, and the omitted integrand is nonnegative. For $N = 20, L = 4, J \in \{0, 2, \dots, 64\}$, all three denominator signs are strictly positive. No correction is required.

1. Exact integrand and the disjoint omitted regions

For $y \geq 0, n \geq 1$, and nonnegative integer J , write

$$x_n(y) = \pi n^2 e^{2y}, \quad f_{J,n}(y) = y^J e^{y/2} (16x_n(y)^2 - 24x_n(y)) e^{-x_n(y)}. \quad (\text{TT1})$$

For $J = 0$, the polynomial y^0 is identically one, including at zero. The finite callback integrates $\sum_{n=1}^{N-1} f_{J,n}$ on $[0, L]$. Its omitted part is exactly

$$\int_0^L \sum_{n=N}^{\infty} f_{J,n}(y) dy + \int_L^{\infty} \sum_{n=1}^{\infty} f_{J,n}(y) dy. \quad (\text{TT2})$$

The shared boundary has measure zero. There is no extra multiplicity or factor of two in this decomposition of the literal half-line integral.

Since $x_n(y) \geq \pi > 3/2$,

$$0 < 16x_n(y)^2 - 24x_n(y) \leq 16x_n(y)^2.$$

Consequently

$$0 \leq f_{J,n}(y) \leq 16\pi^2 n^4 y^J \exp\left(\frac{9}{2}y - \pi n^2 e^{2y}\right). \quad (\text{TT3})$$

The factor $9/2$ is the exact sum $4 + 1/2$ coming from $x_n(y)^2$ and $e^{y/2}$. The negative $-24x_n$ is retained in the finite callback; dropping it only increases the omitted-tail majorant.

2. Omitted n -sum on $0 \leq y \leq L$

On this interval, $y^J \leq L^J$, $e^{9y/2} \leq e^{9L/2}$, and $e^{-\pi n^2 e^{2y}} \leq e^{-\pi n^2}$. Therefore the first term in (TT2) is at most

$$16\pi^2 L^{J+1} e^{9L/2} \sum_{n=N}^{\infty} n^4 e^{-\pi n^2}. \quad (\text{TT4})$$

For $a_n = n^4 e^{-\pi n^2}$, the exact ratio and its majorant satisfy, for every $n \geq N$,

$$\begin{aligned} \frac{a_{n+1}}{a_n} &= (1 + 1/n)^4 e^{-\pi(2n+1)} \\ &\leq e^{4/n - \pi(2n+1)} \leq e^{4/N - 2\pi N} =: \rho_n. \end{aligned} \quad (\text{TT5})$$

The first inequality uses $\log(1 + 1/n) \leq 1/n$. The second drops the additional negative term $-\pi$ and uses $1/n \leq 1/N$ and $n \geq N$. If $\rho_n < 1$, induction gives $a_{N+j} \leq a_N \rho_n^j$, hence

$$0 \leq \int_0^L \sum_{n=N}^{\infty} f_{J,n}(y) dy \leq \frac{16\pi^2 L^{J+1} e^{9L/2} N^4 e^{-\pi N^2}}{1 - e^{4/N - 2\pi N}} =: T_n. \quad (\text{TT6})$$

This is exactly the returned expression `tn`, including its first omitted index N and denominator.

3. Omitted y -interval, including every $n \geq 1$

For fixed $y \geq 0$, let $a_n(y) = n^4 e^{-\pi n^2 e^{2y}}$. For every $n \geq 1$,

$$\begin{aligned} \frac{a_{n+1}(y)}{a_n(y)} &= (1 + 1/n)^4 e^{-\pi(2n+1)e^{2y}} \\ &\leq e^{4/n - 2\pi n e^{2y}} \leq e^{4 - 2\pi e^{2y}} \leq e^{4 - 2\pi} =: \rho_y. \end{aligned} \quad (\text{TT7})$$

Thus

$$\sum_{n=1}^{\infty} n^4 e^{-\pi n^2 e^{2y}} \leq \frac{e^{-\pi e^{2y}}}{1 - \rho_y}.$$

Combining this with (TT3), the second omitted integral in (TT2) is at most

$$\frac{16\pi^2}{1 - \rho_y} \int_L^{\infty} F_J(y) dy, \quad F_J(y) = y^J e^{9y/2 - \pi e^{2y}}. \quad (\text{TT8})$$

For $L > 0$, F_J is positive on $[L, \infty)$. Its logarithmic derivative is exactly

$$\frac{F_J'(y)}{F_J(y)} = \frac{J}{y} + \frac{9}{2} - 2\pi e^{2y} \leq \frac{J}{L} + \frac{9}{2} - 2\pi e^{2L} = -\delta, \quad \delta := 2\pi e^{2L} - \frac{9}{2} - \frac{J}{L}. \quad (\text{TT9})$$

Both inequalities have the displayed direction because $J \geq 0$, $y \geq L > 0$, and the exponential is increasing. When $\delta > 0$, integrating (TT9) from L to y gives $F_J(y) \leq F_J(L) e^{-\delta(y-L)}$. Integrating this inequality proves

$$0 \leq \int_L^{\infty} \sum_{n=1}^{\infty} f_{J,n}(y) dy \leq \frac{16\pi^2 L^J e^{9L/2 - \pi e^{2L}}}{(1 - e^{4-2\pi})(2\pi e^{2L} - 9/2 - J/L)} =: T_y. \quad (\text{TT10})$$

This is exactly ty and its returned decay denominator. The constant geometric majorant ρ_y is deliberately larger than the y -dependent ratio; using it cannot underestimate the tail.

The bounds also justify integrability of the positive series. Tonelli's theorem permits the nonnegative sums and integrals throughout; (TT6) and (TT10), together with the finite continuous integral, show their total is finite.

4. Denominator signs at the actual parameters

For $N = 20$, $L = 4$, and every stated $J \leq 64$, the elementary inequalities $\pi > 3$ and $e^8 > 1 + 8 = 9$ give

$$\begin{aligned} \frac{4}{N} - 2\pi N &< \frac{1}{5} - 120 = -\frac{599}{5}, \\ 4 - 2\pi &< -2, \\ \delta = 2\pi e^8 - \frac{9}{2} - \frac{J}{4} &> 54 - \frac{9}{2} - 16 = \frac{67}{2} > 0. \end{aligned} \quad (\text{TT11})$$

Hence $1 - \rho_n > 1 - e^{-599/5} > 0$, $1 - \rho_y > 1 - e^{-2} > 0$, and $\delta > 67/2$. These are proved lower bounds for the exact denominators at every required even J ; evenness is not needed for the estimates.

The code independently requires the Arb comparisons for those signs to succeed. All quantities in T_n, T_y are nonnegative, and the omitted part lies in $[0, T_n + T_y]$. Therefore using the upper endpoint of the computed enclosure of $\text{tn} + \text{ty}$ and taking its interval hull with zero gives the correct one-sided tail enclosure. This conclusion uses the printed real interval operations and the exact inequalities above; it does not assert that this review executed those operations.

5. Callback analyticity

The finite callback, as a function of complex y , is

$$y^J e^{y/2} \sum_{n=1}^{N-1} (16\pi^2 n^4 e^{4y} - 24\pi n^2 e^{2y}) \exp(-\pi n^2 e^{2y}). \quad (\text{TT12})$$

Every power y^J has a nonnegative integer exponent. All other operations are finite sums, products, and compositions of exponential functions. The function is therefore entire, including at $y = 0$ and for $J = 0$. It uses no conjugation, absolute value, logarithm, division by y , or noninteger power. Accordingly the `analytic=True` callback branch requires no restriction of its complex domain.

The value returned for a complex ball encloses the expression using the real Arb enclosure of π . For each fixed coefficient value within that enclosure, the same finite expression is entire; in particular the target with the exact constant π is entire. Possible nonfinite numerical enclosures do not create an analytic singularity, and the caller separately rejects a nonfinite integration result.

The program partitions $[0, 4]$ into the eight exact half-unit subintervals with endpoints $j/2, (j+1)/2$. The omitted regions and the denominators proved above match that finite integration interval and the loop $n=1,\dots,19$ exactly. This review accepts the tail formulas and callback analyticity without rerunning that integration.

Complete genus-zero product proof

Complete product proof for the original theta entire function

This appendix proves the genus-zero factorization used in the counterfactual workbench. It is a proof of the classical special case, included to make the application's analytic dependencies explicit.

A sublinear growth bound and its zeros

Let F be entire, $F(0) \neq 0$, and suppose that for some $0 < \alpha < 1$ and constants $C, R_0 > 0$,

$$\log \max_{|z| \leq R} |F(z)| \leq CR^\alpha \quad (R \geq R_0). \quad (\text{GF1})$$

The workbench's original F satisfies this with $\alpha = 3/4$: its explicitly proved bound $\log M_F(R) = O(\sqrt{R} \log R)$ implies (GF1) because $\log R/R^{1/4}$ is bounded for $R \geq 1$. Let the nonzero zeros be a_j , repeated with multiplicity, and let $n(r)$ count those with $|a_j| \leq r$.

We first prove the required counting formula. Choose a circle $|z| = R$ containing no zero. There are finitely many zeros in its closed disk, since otherwise an accumulation point would force the nonzero entire function to vanish identically. Divide F by their linear factors; the resulting holomorphic nonvanishing function on a neighborhood of the disk has a holomorphic logarithm there. Indeed its logarithmic derivative is holomorphic on the disk and has a primitive, as follows by termwise integration of its Taylor series on concentric smaller disks. The exponential of that primitive has the same logarithmic derivative; their quotient is constant, giving the logarithm after choosing the constant at zero. The real part of this logarithm has its value at zero as its circle mean, by integrating its Taylor series. For $|a| < R$,

$$\frac{1}{2\pi} \int_0^{2\pi} \log |Re^{it} - a| dt = \log R. \quad (\text{GF2})$$

To see this, expand $\log(1 - ae^{-it}/R) = -\sum_{\ell \geq 1} (a/R)^\ell e^{-i\ell t}/\ell$, which converges uniformly on the circle; each term has zero mean. Taking the logarithm of the removed factors now gives

$$\frac{1}{2\pi} \int_0^{2\pi} \log |F(Re^{it})| dt - \log |F(0)| = \sum_{|a_j| < R} \log \frac{R}{|a_j|}. \quad (\text{GF3})$$

For any sufficiently large r , choose $R \in (2r, 3r)$ containing no zero. All terms are nonnegative and every zero with $|a_j| \leq r$ contributes at least $\log 2$. Equations (GF1)–(GF3) therefore imply $n(r) \leq C_1 r^\alpha$, for a fixed positive C_1 and all sufficiently large r .

Choose a positive r_0 smaller than the modulus of every zero; such a radius exists since $F(0) \neq 0$. After enlarging C_1 , the inequality $n(r) \leq C_1 r^\alpha$ holds for every $r \geq r_0$. The sum over successive dyadic annuli gives

$$\sum_j \frac{1}{|a_j|} \leq \sum_{\ell \geq 0} \frac{n(2^{\ell+1}r_0)}{2^\ell r_0} \leq C_1 2^\alpha r_0^{\alpha-1} \sum_{\ell \geq 0} 2^{(\alpha-1)\ell} < \infty. \quad (\text{GF4})$$

The inequalities can overcount zeros and still give the stated upper bound. If the zero set is empty, every sum is zero.

The product and a lower bound on selected circles

Define the literal product

$$P(z) = \prod_j (1 - z/a_j). \quad (\text{GF5})$$

On a fixed disk $|z| \leq A$, all sufficiently late factors have $|z/a_j| \leq 1/2$. The power series logarithm satisfies $|\log(1 - z/a_j)| \leq 2|z/a_j|$, so (GF4) makes its tail uniformly convergent there. The finite preceding factors and that exponential tail define an entire function, locally uniformly independent of the ordering. Its zeros and their multiplicities are exactly the a_j , and $P(0) = 1$. Thus

$$H(z) = F(z)/P(z) \quad (\text{GF6})$$

extends through every zero to a nonvanishing entire function. An entire primitive of H'/H , obtained by integrating its entire power series, gives an entire logarithm h with $H = e^h$ after adjusting its constant.

We prove h constant without assuming any growth bound for $1/P$ away from the following selected circles. Let t be sufficiently large, put $N_t = n(4t)$, and set $\delta_t = t/[4(N_t + 1)]$. Delete from $[t, 2t]$ each open interval of radius δ_t about a number $|a_j| \leq 4t$. Their total length is at most $2N_t\delta_t < t/2$. Hence a radius $R_t \in [t, 2t]$ remains with $|R_t - |a_j|| \geq \delta_t$ for every such zero. For $|z| = R_t$,

$$|1 - z/a_j| \geq \frac{\delta_t}{4t} = \frac{1}{16(N_t + 1)} \quad (|a_j| \leq 4t). \quad (\text{GF7})$$

For the remaining zeros, $|z/a_j| \leq 1/2$, and the same logarithm estimate used for (GF5) gives $\log|1 - z/a_j| \geq -2|z|/|a_j|$. Another dyadic estimate, retaining only $|a_j| > 4t$, gives

$$\sum_{|a_j| > 4t} \frac{1}{|a_j|} \leq \sum_{\ell \geq 0} \frac{n(2^{\ell+1}4t)}{2^{\ell}4t} \leq C_2 t^{\alpha-1}, \quad (\text{GF8})$$

where $C_2 = C_1 2^{\alpha} 4^{\alpha-1}/(1 - 2^{\alpha-1})$ is positive and fixed. Summing the logarithms of (GF7) and the uniformly convergent tail proves, on the entire selected circle,

$$\log|P(z)| \geq -N_t \log(16(N_t + 1)) - 2R_t C_2 t^{\alpha-1} \geq -C_3 t^{\alpha} \log t \quad (\text{GF9})$$

for another fixed constant $C_3 > 0$ and all large t . This estimate keeps every zero, including repetitions.

The zero-free factor is constant

Combining (GF1), (GF6), and (GF9) shows that $\Re h(z) \leq C_4 t^{\alpha} \log t$ on $|z| = R_t$. Write $h(z) = \sum_{\ell \geq 0} d_{\ell} z^{\ell}$ and put $B_t = C_4 t^{\alpha} \log t$, enlarged if necessary so that the bound holds on every selected circle under consideration. The Taylor series, uniformly convergent on a circle, gives

$$d_{\ell} R_t^{\ell} = \frac{1}{\pi} \int_0^{2\pi} \Re h(R_t e^{i\theta}) e^{-i\ell\theta} d\theta \quad (\ell \geq 1). \quad (\text{GF10})$$

The integral of the constant B_t against $e^{-i\ell\theta}$ is zero. Because $B_t - \Re h \geq 0$ on the circle, its mean and the mean-value identity yield

$$|d_{\ell}| R_t^{\ell} \leq \frac{1}{\pi} \int_0^{2\pi} (B_t - \Re h(R_t e^{i\theta})) d\theta = 2(B_t - \Re h(0)). \quad (\text{GF11})$$

Since $R_t \geq t$ and $\alpha < 1$, the right side divided by R_t^{ℓ} tends to zero for every integer $\ell \geq 1$. Thus every such d_{ℓ} is zero. It follows that H is constant. Evaluating at zero now proves the full factorization

$$F(z) = F(0) \prod_j (1 - z/a_j). \quad (\text{GF12})$$

This also covers finitely many zeros and the empty zero set.

Logarithmic coefficients with their full multiplicities

Group equal zeros in (GF12), writing distinct reciprocals $\beta_j = a_j^{-1}$ and multiplicities m_j . Equation (GF4) becomes $\sum_j m_j |\beta_j| < \infty$. Their supremum is finite. Choose a sufficiently small positive r such that $r|\beta_j| \leq 1/2$ for all j . The derivative series $\sum_j m_j \beta_j / (1 - \beta_j z)$ converges uniformly on $|z| \leq r$, by domination with $2 \sum_j m_j |\beta_j|$. It is the negative logarithmic derivative of (GF12), by termwise differentiation of its locally uniformly convergent logarithm, or by integrating the uniformly convergent derivative series and matching its value at zero. Expanding each denominator and interchanging absolutely convergent sums gives

$$-\frac{F'(z)}{F(z)} = \sum_j \frac{m_j \beta_j}{1 - \beta_j z} = \sum_{n \geq 0} \left(\sum_j m_j \beta_j^{n+1} \right) z^n. \quad (\text{GF13})$$

Every series interchange is justified by $\sum_j m_j |\beta_j| / (1 - r|\beta_j|) < \infty$. Equations (GF12)–(GF13) prove exactly the product and power-sum input used in the workbench and (TC16), with the unchanged $F(0) = g(1/2)$.

Full independent error-control proof

Independent review of the theta error bounds and finite rational witness

Date: 2026-09-13. The complete source `tau_theta_hankel_error_control_20260913.tex`, TC1–TC28, was read at SHA-256 `55f7a9cce6f22b11a30321a7ba726f300a990e78c39780a43b7fdd692c7935e5`. This review independently checks the tail estimates, finite algebra, explicit degree cutoff, coefficient rounding, and interval detection of a negative witness. The integration was not rerun, and the source was not edited.

Finding. The mathematical formulas TC1–TC26 are consistent with the retained source identities, and TC23 and TC25 give the claimed margins, including their endpoint cases. The finite positive-definiteness implication in TC27 follows from the stated exact pivot certificates; their complete numerical endpoint replay is the separate parent review. The displayed TC28 decimal enclosure matches its display string in `certificate_dimension16_1024.json`.

One prose correction was reported: “No coefficient is removed by the rounding rule” can be read as asserting that a nonzero coefficient never rounds to zero, which is false. The precise statement is: every original coefficient slot is included in the error estimate; nearest-grid rounding may produce zero. The theorem should explicitly specify rounding to a nearest multiple of $1/H$, with either choice allowed at a tie. These changes clarify the already used bound $1/(2H)$; no displayed inequality needs alteration.

1. Retained integral and complete positive tail bounds

The review uses the workbench’s original reflection and Mellin identities stated before TC2, with $g = 2\xi$. There are two distinct factors of two: one in $f_0 = 2 \sum \phi_*$, and one from reflection of the even moment. Their product gives the factor four in TC2. Expanding

$$y^J e^{y/2} (16x_n^2 - 24x_n) e^{-x_n}, \quad x_n = \pi n^2 e^{2y},$$

gives exactly

$$4y^J (4\pi^2 n^4 e^{9y/2} - 6\pi n^2 e^{5y/2}) e^{-\pi n^2 e^{2y}}. \quad (\text{TR1})$$

Thus TC2 and the callback TC9 agree with their original constants.

For $y \geq 0, n \geq 1, x_n \geq \pi > 3/2$, so

$$0 \leq y^J e^{y/2} (16x_n^2 - 24x_n) e^{-x_n} \leq 16\pi^2 n^4 y^J e^{9y/2 - \pi n^2 e^{2y}}. \quad (\text{TR2})$$

For the polynomial y^0 , its value at zero is one. For $J > 0$, the vanishing factor at zero creates no exception to the inequality.

For $n = N + \ell, \ell \geq 0$, the elementary bounds $(N + \ell)^4 \leq N^4 e^{4\ell/N}$ and $(N + \ell)^2 \geq N^2 + 2N\ell$ imply

$$\sum_{n=N}^{\infty} n^4 e^{-\pi n^2 e^{2y}} \leq \frac{N^4 e^{-\pi N^2 e^{2y}}}{1 - e^{4/N - 2\pi N e^{2y}}} \leq \frac{N^4 e^{-\pi N^2}}{1 - r_N}. \quad (\text{TR3})$$

Here $r_N = e^{4/N - 2\pi N} \in (0, 1)$: $4/N - 2\pi N \leq 4 - 2\pi < 0$ for $N \geq 1$. Multiplication by (TR2), bounding $y^J \leq L^J$ and $e^{9y/2} \leq e^{9L/2}$, and integrating over an interval of length L prove exactly

$$T_{J,N,L} = \frac{16\pi^2 L^{J+1} e^{9L/2} N^4 e^{-\pi N^2}}{1 - r_N}. \quad (\text{TR4})$$

For the omitted interval $y > L$, use (TR3) with $N = 1$, retaining its exponential numerator. This gives

$$I_J(y) \leq \frac{16\pi^2}{1 - r_1} y^J e^{9y/2 - \pi e^{2y}}, \quad r_1 = e^{4 - 2\pi}. \quad (\text{TR5})$$

For $y = L + v, v \geq 0$, the inequalities $(L + v)^J \leq L^J e^{Jv/L}$ and $e^{2(L+v)} \geq e^{2L}(1 + 2v)$ yield

$$y^J e^{9y/2 - \pi e^{2y}} \leq L^J e^{9L/2 - \pi e^{2L}} e^{-(2\pi e^{2L} - 9/2 - J/L)v}. \quad (\text{TR6})$$

Integrating when $\lambda_J(L) = 2\pi e^{2L} - 9/2 - J/L > 0$ proves exactly

$$Y_{J,L} = \frac{16\pi^2 L^J e^{9L/2 - \pi e^{2L}}}{(1 - r_1) \lambda_J(L)}. \quad (\text{TR7})$$

The first tail covers $0 \leq y \leq L, n \geq N$, and the second covers $y > L, n \geq 1$; their interiors are disjoint. The positive integrands permit Tonelli before integrability is established. The finite bounds then prove convergence and TC4.

At $N = 20, L = 4, J \leq 64$, the exact denominator signs follow already from $\pi > 3, e^8 > 41$:

$$\frac{4}{N} - 2\pi N < -\frac{599}{5}, \quad 4 - 2\pi < -2, \quad \lambda_J(4) > 246 - \frac{9}{2} - 16 = \frac{451}{2} > 0. \quad (\text{TR8})$$

Thus the quoted actual-parameter sign check is correct.

For fixed J , first choose L large enough to make $Y_{J,L}$ arbitrarily small; its double-exponential decay dominates the polynomial and exponential factors. For that fixed L , increase N to make $T_{J,N,L}$ arbitrarily small. This is a sequential choice of L , then N , not a claim that the first tail tends to zero with L while N remains fixed.

The finite integrand is entire in complex y : y^J has a nonnegative integer exponent, and all remaining operations are finite sums, products, and compositions of exponentials. There are no branch cuts, absolute values, or conjugations in the callback. Its analyticity therefore meets the recorded analytic-ball callback contract, including at $y = 0, J = 0$.

2. Recurrence, interval propagation, and finite matrix signs

The retained entire-function identity $F(z^2) = g(1/2 + iz)$, together with the even moments, gives $a_j = (-1)^j M_{2j}/(2j)!$. Extracting the coefficient of w^n from

$$F(w) \sum_{n \geq 0} b_n w^n = -F'(w)$$

gives

$$a_0 b_n + \sum_{j=1}^n a_j b_{n-j} = -(n+1)a_{n+1}, \quad (\text{TR9})$$

which is TC11. The original $a_0 = M_0 > 0$ is retained in this recurrence.

Each interval operation encloses its exact operation. Induction on n , with a positive lower bound for a_0 , therefore encloses every exact b_n . Repeated use of an interval variable can increase enclosure width, but cannot invalidate containment.

For the exact symmetric matrix, the recursion TC13 successively constructs a unit lower triangular L and diagonal D . At stage j , the formula for L_{ij} gives

$$A_{ij} = \sum_{k < j} L_{ik} D_k L_{jk} + L_{ij} D_j \quad (i > j),$$

and the pivot formula gives the same equality for $i = j$, since $L_{jj} = 1$. Symmetry supplies the other half. Thus $A = LDL^T$. If all interval pivots have positive lower endpoints, induction proves that the exact pivots are positive and every division is defined. Hence for $v \neq 0$

$$v^* A v = \sum_j D_j |(L^T v)_j|^2 > 0. \quad (\text{TR10})$$

The vector is nonzero after the invertible triangular map. This proves the stated positive-definiteness implication.

For a real rational vector c , let $\gamma_n = \sum_{r+s=n} c_r c_s$. Expanding the complete quadratic form gives

$$c^T H_d c = \sum_{n=0}^{2d} \gamma_n b_n. \quad (\text{TR11})$$

This includes off-diagonal cross terms twice in the coefficient convolution when their indices differ. If $b_n \in [\ell_n, u_n]$, choosing u_n when $\gamma_n \geq 0$ and ℓ_n when $\gamma_n < 0$ maximizes that term. Therefore TC15 is a valid rational upper bound. No square-modulus form is substituted for this real bilinear form.

3. Reciprocal-zero map and the full polynomial form

The source imports the workbench's absolutely summable reciprocal-zero expansion TC16. All deductions below retain those exact multiplicities and its full infinite tail. The coordinate map in TC17 is verified directly:

$$-\frac{1}{(\delta + i\gamma)^2} = \frac{\gamma^2 - \delta^2 + 2i\gamma\delta}{(\gamma^2 + \delta^2)^2}. \quad (\text{TR12})$$

Replacing ρ by $1 - \rho$ changes $\rho - 1/2$ to its negative and leaves the square unchanged. At a zero, $\rho - 1/2 \neq 0$, because the source value $g(1/2) = M_0$ is positive. The map $w = -(\rho - 1/2)^2$ has nonzero derivative there. It preserves the local zero order, so the two reflected source zeros correspond to one zero of F of that same order. This is the exact multiplicity map used in TC16–TC17.

The finiteness of $S_* = \sum_j m_j |\beta_j|$ implies $\sup_j |\beta_j| < \infty$, since $m_j \geq 1$. Put $R_* = \max\{1, \sup_j |\beta_j|\}$. For $P(X) = \sum p_r X^r$,

$$|P(\beta_j)| \leq \sum_r |p_r| R_*^r = \|P\|_{R_*}. \quad (\text{TR13})$$

Thus $\mathcal{Q}(P) = \sum_j m_j \beta_j P(\beta_j)^2$ converges absolutely. It is real for real P because the reciprocal set and multiplicities are conjugation stable. Finite expansion of P^2 gives TC19 exactly.

For real polynomials P, E , the identity $(P + E)^2 - P^2 = 2PE + E^2$, applied termwise in that absolutely convergent series, gives

$$|\mathcal{Q}(P + E) - \mathcal{Q}(P)| \leq S_* \|E\|_{R_*} (2\|P\|_{R_*} + \|E\|_{R_*}). \quad (\text{TR14})$$

This verifies TC20 without dropping any reciprocal zero or multiplicity.

4. Separator, explicit cutoff, and all edge cases

Fix a nonreal reciprocal zero β , its conjugate, and their common multiplicity m . Let $r = |\beta|/2 > 0$. In TC21, “all other” excludes both selected points. Every element of $\mathcal{A}$ contributes at least r to S_* , so $\#\mathcal{A} \leq S_*/r$ and the set is finite. It is conjugation stable. Therefore B has real coefficients and $B(\beta) \neq 0$. The empty product is one if $\mathcal{A}$ is empty.

Either choice of c with $\beta c^2 = -1$ is nonzero. Set w_N, u_N, v_N as in TC24. They satisfy

$$u_N + v_N \beta = \Re w_N - (\Re \beta) v_N + v_N (\Re \beta + i \Im \beta) = w_N. \quad (\text{TR15})$$

Consequently $P_N(\beta) = c$, and real coefficients give $P_N(\bar{\beta}) = \bar{c}$. The two selected summands of $\mathcal{Q}(P_N)$ are each exactly $-m$. Every other reciprocal zero of modulus at least r is a root of B , so its entire weighted summand is zero, including its multiplicity.

For $|X| \leq r$, the literal product gives $|B(X)| \leq B_0$. Also

$$\begin{aligned} |u_N + v_N X| &\leq |\Re w_N| + (|\Re \beta| + r) |v_N| \\ &\leq |w_N| \left(1 + \frac{|\Re \beta| + r}{|\Im \beta|} \right). \end{aligned} \quad (\text{TR16})$$

Multiplication by $|X|^N |B(X)|$, and using $|w_N| = |c|/(|\beta|^N |B(\beta)|)$, gives

$$|P_N(X)| \leq C_* 2^{-N}. \quad (\text{TR17})$$

This includes $N = 0$, with X^0 the constant polynomial one. The full remaining reciprocal tail therefore has absolute contribution at most $C_*^2 T_* 4^{-N}$, yielding

$$\mathcal{Q}(P_N) \leq -2m + C_*^2 T_* 4^{-N}. \quad (\text{TR18})$$

If $T_* = 0$, the last term is zero for every $N \geq 0$. If $T_* > 0$, set $A = C_*^2 T_*/m > 0$ and $x = \log A / \log 4$. When $x \geq 0$, $1 + \lfloor x \rfloor > x$, so every $N \geq N_*$ satisfies $N > x$. When $x < 0$, $N \geq 0 > x$, regardless of whether the inner integer in TC23 is zero or negative. Hence in both cases

$$A 4^{-N} < 1, \quad C_*^2 T_* 4^{-N} < m. \quad (\text{TR19})$$

In particular TC23 gives the claimed weak bound $\mathcal{Q}(P_N) \leq -m$. At $A = 4^s$ with integer $s \geq 0$, the chosen cutoff is $s + 1$, not s ; the boundary case is therefore safe. The formula is conservative by one degree at that boundary, and is mathematically valid as printed.

5. Rational coefficient accuracy and interval detection

The polynomial P_N has real coefficients and degree at most $D = N + \#\mathcal{A} + 1$. Retain all $D + 1$ coefficient slots, inserting exact zeros above its actual degree if necessary. For a positive integer H , round each coefficient to a

nearest element of $H^{-1}\mathbb{Z}$. Either nearest element is allowed at a tie. The error polynomial $E = \tilde{P}_N - P_N$ then has

$$|e_r| \leq \frac{1}{2H}, \quad \eta := \|E\|_{R_*} \leq \frac{\sum_{r=0}^D R_*^r}{2H}. \quad (\text{TR20})$$

Every denominator in TC25 is positive: $m > 0$, $S_* > 0$ because it includes the selected pair, and $2\|P_N\|_{R_*} + 1 > 0$. The finite geometric sum is positive. Thus a sufficiently large positive integer H always satisfies TC25.

The two bounds in TC25 give $\eta \leq 1$ and $\eta \leq m/[2S_*(2\|P_N\|_{R_*} + 1)]$. By (TR14),

$$\begin{aligned} |\mathcal{Q}(\tilde{P}_N) - \mathcal{Q}(P_N)| &\leq S_*\eta(2\|P_N\|_{R_*} + \eta) \\ &\leq S_*\eta(2\|P_N\|_{R_*} + 1) \leq m/2. \end{aligned} \quad (\text{TR21})$$

Together with (TR18)–(TR19), this proves $\mathcal{Q}(\tilde{P}_N) \leq -m/2 < 0$. The error estimate includes a coefficient that rounds from a nonzero value to zero; such rounding is permitted and does not invalidate the proof.

For clarity, the eventual finite-interval claim has a direct quantitative margin. Write c for the rational coefficient vector of $\tilde{P}_N$, define γ_n as in (TR11), and set $A_c = \sum_n |\gamma_n|$. The polynomial is nonzero because its form is negative, so its square is nonzero and $A_c > 0$. If rational enclosing intervals for every required b_n have width at most $m/(4A_c)$, then their signed upper bound satisfies

$$U(c) - \mathcal{Q}(\tilde{P}_N) \leq \sum_n |\gamma_n|(u_n - \ell_n) \leq m/4, \quad U(c) \leq -m/4 < 0. \quad (\text{TR22})$$

This proves an explicit sufficient finite enclosure width. There is no need to assume cancellation or independent errors.

For finitely many required coefficients, shrinking the moment intervals shrinks the resulting b_n intervals: TC11 uses finitely many additions, products, and reciprocals, and one can first restrict the a_0 interval to have lower endpoint at least $M_0/2 > 0$. Every such interval operation is continuous at those exact inputs; induction in n proves convergence of enclosure widths to zero. The positive tail bounds can be made arbitrarily small by the sequential choice of L, N established above.

The finite integral itself admits arbitrarily small validated errors. One direct existence proof uses the entire finite integrand f on $[0, L]$. On all unit complex discs about that interval, a finite bound for $|f|$ is

$$K = (L+2)^J e^{(L+1)/2} \sum_{n=1}^{N-1} (16\pi^2 n^4 e^{4(L+1)} + 24\pi n^2 e^{2(L+1)}) e^{\pi n^2 e^{2(L+1)}}. \quad (\text{TR23})$$

Indeed $\Re y \leq L+1$, $|y| \leq L+2$, and $|e^{-\pi n^2 e^{2y}}| \leq e^{\pi n^2 e^{2\Re y}}$. Cauchy's derivative estimate on each unit circle gives $|f'(x)| \leq K$ for $x \in [0, L]$. Composite midpoint quadrature on a uniform mesh of width $h = L/s$ has error at most $KLh/4$, by integrating $K|x - x_{\text{mid}}|$ on each interval. Finite evaluations of the explicit exponential expression can be enclosed to arbitrary accuracy; outward rational bounds for K suffice. If $N = 1$, the finite integrand is zero and this step is immediate. This proves the finite integration approximation needed for (TR22), independently of any fixed evaluation limit of the particular integration run.

6. Relation to the recorded finite certificate

For dimension 16, $d = 15$. The unshifted matrix uses $b_0, \dots, b_{30}$, and the shifted one uses $b_1, \dots, b_{31}$. TC11 computes these from $a_0, \dots, a_{32}$, hence exactly the stated 33 even moments $M_0, M_2, \dots, M_{64}$. The index ranges are consistent. Positivity of both sets of sixteen exact pivots proves TC27 by (TR10); every smaller leading matrix is positive definite by restriction of the same positive quadratic form.

The parent performs the complete rational endpoint and replay audit. This subreview supplies the independent mathematical derivation and does not turn a finite-dimensional sign result into a claim at larger dimensions. Likewise the separator theorem constructs and bounds the witness associated with each specified nonreal reciprocal zero; it asserts no universal cutoff and supplies no nonreal zero of the actual zeta function.

Final mathematical and exact-rational acceptance

Root acceptance of the original theta certificate and complete proof closure

13 September 2026. This is the final local mathematical and execution review for the current finite certificate. It does not close the wider Split-Zero/RH research goal.

Exact accepted calculation

Both original real symmetric matrices

$$H_{15} = (b_{i+j})_{0 \leq i, j \leq 15}, \quad H_{15}^{(1)} = (b_{i+j+1})_{0 \leq i, j \leq 15}$$

are positive definite. The coefficients are those of the actual $-F'/F$, with $F(z^2) = 2\xi(1/2 + iz)$. All 33 moments $M_0, M_2, \dots, M_{64}$ are calculated from the original theta integral. All 32 pivots and all 32 leading determinants have positive rigorous lower endpoints. The original mass and all monomial coordinates remain unchanged.

The complete integrator, CH.1–19 method proof, TR1–33 analytic/code review, and TC1–28 root proof have been read. The two omitted regions are disjoint, their integrands are nonnegative, and their upper bounds retain the factor $16\pi^2$, the original exponential factors and both exact denominators. The complex integration callback is entire. Its real ball contains the real integral; the positive tail interval is added with outward rounding. The dyadic exporter uses exact outward lower/upper endpoints and their integer mantissa/exponent, with no decimal conversion in the certificate.

Independent integer arithmetic and actual replay

The saved independent script `theta_certified_hankel_exact_rational_review_20260913.py` was read completely. It imports no flint or floating-point arithmetic. It uses a fixed grid $Q = 2^{4096}$: an interval is the exact pair $[\ell/Q, u/Q]$ of integers. Addition and negation are exact; multiplication takes the minimum and maximum of all four integer endpoint products and rounds outward after division by Q . Squaring uses zero as its lower endpoint when the interval crosses zero. For a positive interval its reciprocal is enclosed by

$$\left[\frac{\lfloor Q^2/u \rfloor}{Q}, \frac{\lceil Q^2/\ell \rceil}{Q} \right].$$

Negative reciprocals are obtained by the exact negation map. These formulas prove containment of every operation, including all rounded operations.

The script reconstructs each moment from the eight exported finite-piece intervals and the sum of the two exported tail upper bounds. It does not use the saved moment, coefficient or pivot values to calculate its result. It then propagates the original factorial/sign formula and logarithmic recurrence and independently performs both unscaled LDL recursions. Every denominator is checked positive before division. The complete matrix identity and positive-definiteness proof are TC13 and CH.15–17.

Root ran a fresh validated integration after the independent review worker hit its usage limit. The fresh certificate is `reviewer_replay_dimension16_1024.json`, SHA-256 `0fcac21e20680a6d6a3fab1671b0782b56de005ce749ee33373b1193f20a0af7`. The complete numerical payload agrees exactly with the original certificate; elapsed time and platform fields are excluded from that comparison. Root then executed the independent integer script successfully in ordinary Python. It validated 1,400 exact intervals, reconstructed both matrix certificates, and proved all 32 positive pivots and determinants. It also proved by exact rational comparison that the readable TC28 interval contains the full exported determinant interval.

The original certificate SHA-256 is `869ebacd3f82e46c3fe5fac1f07bcbaae7f4c333e1a5ba2e0ffcb1f91884a0fc`. The reviewed integrator SHA-256 is `a97486edc2fc1f05f155a9539ad280433be39a8f6924efbc8c1c1c4d71033950`. The independent integer script SHA-256 is `a181ef220578eaf232b3e01398d61b1a6f36625f056c5e6bd365453b9ebfa885`. Its successful complete rational output is supplied alongside this proof. The script uses Python assertions for validation and must be run without the optimization switch. No optimized-mode verifier claim is made.

Written-proof corrections and closure

The complete TC mathematical subreview accepted every displayed bound and reported two prose clarifications: coefficient rounding must mean a nearest multiple, with either tie choice allowed; every coefficient slot enters the error bound even when a nonzero coefficient rounds to zero. Both are now explicit in the accepted source. Its current SHA-256 is 8dab2193a7ef3287f58ff5038fc34988883d8899dc4a2a9868098856b48772a5. The earlier subreview remains a record of the reviewed predecessor and its resolved findings. The bounds and their indices were unchanged.

The complete workbench NOTE.tex and WBR1–41 independent proof were read. The archived separator defines u_N and then writes ν_N in its boxed formula. The exact correction is the identity $\nu_N := u_N$; the archive is preserved and the consistent full formula appears in WBR16 and TC24. The rational functional calculus has the exact full local inverse WBR29, and its two reflected modules remain separate through WBR33. The half-trace, nilradical and critical-observation defect are completely proved in WBR34–40.

The workbench originally invokes the classical genus-zero factorization without including its proof. The additional complete source tau_genus_zero_product_proof_20260913.tex, GF1–13, now proves the precise case used. Root wrote and read back that whole proof. It derives the circle-mean zero formula, the $O(r^\alpha)$ zero count, absolute reciprocal summability, uniform product convergence, a lower bound for the product on explicitly selected zero-avoiding circles, and constancy of the entire zero-free factor from its Fourier coefficients. It then derives the full logarithmic-derivative expansion. Taking $\alpha = 3/4$ applies it to WBR9's explicit theta growth bound. Its SHA-256 is 9bade2c098f1b03ba4ec4e526f4dffe4ced36f70bd7a533931a16335b162c394. This is an included proof of a classical theorem, not a novelty claim. It is root-reviewed; no separate independent acceptance of GF is claimed.

The finite dimension-16 certificate does not require factorization or any assumed zero location. Factorization belongs to the separate proof that positivity of the original unshifted matrices in every dimension is equivalent to RH. The complete finite tails and rational rounding formula give rigorous finite witnesses without choosing a replacement metric. The actual computation produced positive finite matrices and no negative witness. It proves no sign statement at larger dimensions.

Final independent product, growth and theta-tail proof

Independent final review: the full genus-zero proof and the positive theta tails

Date: 2026-09-13.

The complete files `tau_genus_zero_product_proof_20260913.tex` (GF1–GF13) and `theta_certified_tail_subreview_20260913.md` (TT1–TT12) were read and checked. The original workbench `NOTE.tex` was also read in full to check the exact function, coefficient convention, and invocation of the product theorem. A separate reviewer read the entire retained `certify_theta_hankel.py` and matched its callback and tail expressions to TT1–TT12. No source file was modified and no integration, Lean run, or remote publication was performed in this review.

Acceptance: GF1–GF13 and TT1–TT12 are mathematically valid with their original constants, signs, multiplicities, and domains. No correction is required. The explicit derivation below supplies the original-theta growth estimate invoked in the first paragraph of GF, and the remainder checks every step of the product argument and of the two tail estimates. The finite computation’s exact arithmetic and numerical integration certification are outside this review; accepting the tail formulas does not by itself certify those separate operations.

1. Exact sources and the original analytic input

The reviewed bytes are:

Source	Bytes	SHA-256
<code>tau_genus_zero_product_proof_20260913.tex</code>	7008	9bade2c098f1b03ba4ec4e526f4dffe4ced36f70bd7a533931a16335b162c394
<code>theta_certified_tail_subreview_20260913.md</code>	6761	4e5c6ad421b6d893e551c1746e05a08be3bf3b0753fae69c16feb5b7025dedf
<code>certify_theta_hankel.py</code>	7472	a97486edc2fc1f05f155a9539ad280433be39a8f6924efbc8c1c1c4d71033950
Original workbench <code>NOTE.tex</code>	20408	df7854f73ecb396c21165f8c032722a851a2a4c893f2553c6da96f88014b62f6

The original source in the workbench is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad f_0(x) = 2 \sum_{n \geq 1} \phi_*(nx), \quad \psi(y) = e^{y/2} f_0(e^y). \quad (\text{GFR1})$$

The supplied Mellin identity and reflection give

$$g(1/2 + iz) = \int_{\mathbb{R}} \psi(y) e^{izy} dy, \quad \psi(-y) = \psi(y), \quad g = 2\xi. \quad (\text{GFR2})$$

The workbench’s coefficients specify the entire function by

$$F(w) = \sum_{j \geq 0} \frac{(-1)^j M_{2j}}{(2j)!} w^j, \quad M_{2j} = \int_{\mathbb{R}} y^{2j} \psi(y) dy, \quad F(z^2) = g(1/2 + iz). \quad (\text{GFR3})$$

All factors of two in (GFR1)–(GFR3) are retained. In particular, the original value is $F(0) = g(1/2) = \int_{\mathbb{R}} \psi(y) dy > 0$.

Here is a completely explicit route from the source to GF1. Put

$$\rho = \exp(4 - 2\pi), \quad C_\theta = \frac{8\pi^2}{1 - \rho}. \quad (\text{GFR4})$$

Because $\pi > 3$, $0 < \rho < 1$. For $n \geq 1$,

$$\frac{(n+1)^4 e^{-\pi((n+1)^2-1)}}{n^4 e^{-\pi(n^2-1)}} = (1 + 1/n)^4 e^{-\pi(2n+1)} \leq e^{4/n-2\pi n} \leq \rho.$$

The first term of the sequence is one, so its sum is at most $(1 - \rho)^{-1}$. For $y \geq 0$, the literal summand of ψ is positive and bounded above by its positive fourth-power term. Consequently

$$\begin{aligned} 0 < \psi(y) &\leq 8\pi^2 e^{9y/2 - \pi e^{2y}} \sum_{n \geq 1} n^4 e^{-\pi(n^2 - 1)e^{2y}} \\ &\leq C_\theta e^{9y/2 - \pi e^{2y}}. \end{aligned} \quad (\text{GFR5})$$

Reflection gives the same estimate with $|y|$. For every fixed $s \geq 0$, this bound times $e^{s|y|}$ is integrable. It justifies the power-series expansion and differentiations in (GFR2) by domination on every compact set of z . The evenness of ψ removes exactly the odd coefficients. It also proves convergence of (GFR3) on every compact set of w , since one can compare the absolute series with the even exponential series at $|z| = \sqrt{|w|}$.

For $R \geq 1$ and every $|w| \leq R$, choose one number z with $z^2 = w$. This pointwise choice uses no holomorphic square-root branch; it only gives $|z| \leq \sqrt{R}$. Let $a = 9/4 + \sqrt{R}/2$. Equation (GFR5) gives

$$\begin{aligned} |F(w)| &\leq 2C_\theta \int_0^\infty e^{(9/2 + \sqrt{R})y - \pi e^{2y}} dy \\ &= C_\theta \int_1^\infty t^{a-1} e^{-\pi t} dt \leq C_\theta \pi^{-a} \Gamma(a). \end{aligned} \quad (\text{GFR6})$$

The substitution is exactly $t = e^{2y}$, so $dy = dt/(2t)$. The factor two from the two half-lines cancels this denominator two. For $a > 1$, the positive function $t^{a-1} e^{-t/2}$ has its maximum at $t = 2(a - 1)$; this follows from its logarithmic derivative $(a - 1)/t - 1/2$. Thus

$$\Gamma(a) = \int_0^\infty (t^{a-1} e^{-t/2}) e^{-t/2} dt \leq 2 \left(\frac{2(a-1)}{e} \right)^{a-1}. \quad (\text{GFR7})$$

Write $K_0 = \max\{0, \log(2C_\theta)\}$. Since $a - 1 = 5/4 + \sqrt{R}/2 \leq 7\sqrt{R}/4$, equations (GFR6)–(GFR7) imply

$$\log M_F(R) \leq K_0 + \frac{7}{4} \sqrt{R} \log \frac{7}{2} + \frac{7}{8} \sqrt{R} \log R. \quad (\text{GFR8})$$

In deriving this upper bound, the terms $-(a - 1) - a \log \pi$ were discarded only because they are negative; the original function and its mass remain unchanged. For $R \geq 1$, $\log R/R^{1/4} \leq 4/e$: putting $u = \log R \geq 0$, the derivative of $ue^{-u/4}$ is $e^{-u/4}(1 - u/4)$, whose maximum value is $4/e$. Hence a concrete choice in GF1 is

$$\alpha = \frac{3}{4}, \quad R_0 = 1, \quad C = K_0 + \frac{7}{4} \log \frac{7}{2} + \frac{7}{2e} > 0. \quad (\text{GFR9})$$

This derivation proves the growth hypothesis for precisely the original F . It does not use zero ordinates, a product formula, or an assumed location of zeros.

2. Counting all zeros, with their multiplicities

For the general entire function in GF1, $F(0) \neq 0$ ensures that the function is not identically zero and that all zeros have positive modulus. Zeros in any compact disk are finite, counting multiplicities. Choose a radius R whose circle has no zero and divide F by the factors $z - a_j$ for all its zeros in that disk. The quotient extends holomorphically through those zeros and is nonvanishing on a slightly larger disk: the boundary circle is compact and nonvanishing, so an annular neighborhood of it contains no zero. Its logarithmic derivative has a primitive on this disk, obtained from its power series. If L' is that logarithmic derivative, differentiating the quotient times e^{-L} gives zero, so this quotient equals a nonzero constant times e^L . A choice of a complex logarithm of the constant supplies its holomorphic logarithm.

The mean of the real part of a holomorphic logarithm equals its central value by its uniformly convergent Taylor series on the circle. For a removed zero $|a_j| < R$, the uniformly convergent expansion

$$\log(1 - a_j e^{-i\theta}/R) = - \sum_{k \geq 1} \frac{(a_j/R)^k e^{-ik\theta}}{k}$$

has real part of mean zero. Thus the mean of $\log |Re^{i\theta} - a_j|$ is exactly $\log R$. Subtracting the central value of the removed linear factors gives GF3, including the term $\log |F(0)|$ and each repeated zero.

For sufficiently large r , the set of excluded radii in $(2r, 3r)$ is finite. Choose any other R there. Every zero with $|a_j| \leq r$ contributes at least $\log 2$, whereas every other term in GF3 is nonnegative. Therefore

$$n(r) \log 2 \leq C(3r)^\alpha - \log |F(0)|. \quad (\text{GFR10})$$

This proves $n(r) \leq C_1 r^\alpha$ for all sufficiently large r . Choose $r_0 > 0$ strictly below the modulus of every zero. Such a radius exists by nonvanishing at zero; if the zero set is empty any positive radius works. On the remaining bounded interval, $n(r)/r^\alpha$ is bounded above by a fixed finite count divided by r_0^α . Enlarging C_1 therefore proves the estimate for every $r \geq r_0$.

Partition the zeros into the half-open annuli $2^k r_0 < |a_j| \leq 2^{k+1} r_0$, $k \geq 0$. There are no zeros omitted below or at r_0 . Each term in the k -th annulus is at most $(2^k r_0)^{-1}$, and that annulus contains at most $n(2^{k+1} r_0)$ zeros. Consequently

$$\sum_j |a_j|^{-1} \leq C_1 2^\alpha r_0^{\alpha-1} \sum_{k \geq 0} 2^{k(\alpha-1)} < \infty. \quad (\text{GFR11})$$

This verifies GF4 without a density assumption and with every multiplicity included.

3. Construction of the literal product and the selected-circle bound

Let $P(z) = \prod_j (1 - z/a_j)$. For a fixed disk $|z| \leq A$, only finitely many zeros have modulus at most $2A$. For every remaining factor, the power-series logarithm obeys

$$|\log(1 - z/a_j)| \leq \sum_{k \geq 1} |z/a_j|^k / k \leq \frac{|z/a_j|}{1 - |z/a_j|} \leq 2A/|a_j|. \quad (\text{GFR12})$$

By (GFR11) these logarithms have an absolutely and uniformly convergent sum. Its exponential has no zero on the disk. Multiplication by the preceding finite factors therefore gives a holomorphic function with exactly the prescribed zeros and orders there. The constructions agree on intersections because they use the same absolutely convergent tail and finite factors. They define the entire P , independent of the enumeration of zeros, with $P(0) = 1$.

At each zero, F and P have the same order and a nonzero leading coefficient, so $H = F/P$ extends as a nonvanishing holomorphic function. Thus H is nonvanishing entire. Integrating the entire power series of H'/H and choosing the constant at zero gives an entire h satisfying $H = e^h$. No bound on $1/P$ has been used in this construction.

For sufficiently large t , set $N_t = n(4t)$ and $\delta_t = t/[4(N_t + 1)]$. The union of the open intervals of radius δ_t about the at most N_t zero moduli at most $4t$ has total length at most $2N_t \delta_t = tN_t/[2(N_t + 1)] < t/2$. The interval $[t, 2t]$ has length t , so a radius R_t outside the union exists. For every $|z| = R_t$ and every zero $|a_j| \leq 4t$, the reverse triangle inequality yields

$$|1 - z/a_j| = \frac{|a_j - z|}{|a_j|} \geq \frac{|R_t - |a_j||}{4t} \geq \frac{1}{16(N_t + 1)}. \quad (\text{GFR13})$$

Repeated zeros merely repeat factors; counting their excluded intervals more than once only weakens the estimate on the union length. For every remaining zero, $|a_j| > 4t$ and $|z| \leq 2t$, so $|z/a_j| < 1/2$. Equation (GFR12) implies $\log |1 - z/a_j| \geq -2|z|/|a_j|$. Using annuli $2^k 4t < |a_j| \leq 2^{k+1} 4t$, exactly the counting argument above gives

$$\sum_{|a_j| > 4t} |a_j|^{-1} \leq \frac{C_1 2^\alpha 4^{\alpha-1}}{1 - 2^{\alpha-1}} t^{\alpha-1} = C_2 t^{\alpha-1}. \quad (\text{GFR14})$$

Both the finite log sum and its uniformly convergent infinite tail can now be evaluated on the full circle, giving

$$\log |P(z)| \geq -N_t \log(16(N_t + 1)) - 2R_t C_2 t^{\alpha-1}. \quad (\text{GFR15})$$

For completeness, put $A = C_1 4^\alpha$. For $t \geq e$, $N_t \leq At^\alpha$, $N_t + 1 \leq (A + 1)t^\alpha$, and $R_t \leq 2t$. Thus

$$\begin{aligned} N_t \log(16(N_t + 1)) &\leq At^\alpha [\log(16(A + 1)) + \alpha \log t], \\ 2R_t C_2 t^{\alpha-1} &\leq 4C_2 t^\alpha. \end{aligned}$$

Taking $C_3 = A[\log(16(A + 1)) + \alpha] + 4C_2$ proves GF9 for all these sufficiently large t . The proof covers infinitely many, finitely many, or no zeros; an empty finite log sum contributes zero.

4. The exact argument forcing the zero-free quotient to be constant

On the selected circle, F and P are nonzero. Therefore $\Re h = \log |H| = \log |F| - \log |P|$. For large t , GF1 applies at R_t and yields

$$\Re h(z) \leq C 2^\alpha t^\alpha + C_3 t^\alpha \log t \leq B_t := (C 2^\alpha + C_3) t^\alpha \log t \quad (|z| = R_t), \quad (\text{GFR16})$$

where $t \geq e$. The product $C 2^\alpha$ retains the original growth constant C ; the tail constant C_2 has already entered through C_3 .

Write $h(z) = \sum_{k \geq 0} d_k z^k$. On any fixed circle this series and its conjugate converge uniformly. For every integer $\ell \geq 1$, multiplying

$$\Re h(R_t e^{i\theta}) = \frac{1}{2} \sum_{k \geq 0} d_k R_t^k e^{ik\theta} + \frac{1}{2} \sum_{k \geq 0} \overline{d_k} R_t^k e^{-ik\theta}$$

by $e^{-i\ell\theta}$ and integrating gives exactly

$$\frac{1}{\pi} \int_0^{2\pi} \Re h(R_t e^{i\theta}) e^{-i\ell\theta} d\theta = d_\ell R_t^\ell. \quad (\text{GFR17})$$

Only the first sum's term $k = \ell$ survives. In particular, the coefficient $1/\pi$ in GF10 is correct. The integral of the constant B_t times $e^{-i\ell\theta}$ is zero. Since $B_t - \Re h \geq 0$, the triangle inequality and the mean-value identity give

$$|d_\ell| R_t^\ell \leq \frac{1}{\pi} \int_0^{2\pi} (B_t - \Re h(R_t e^{i\theta})) d\theta = 2(B_t - \Re h(0)). \quad (\text{GFR18})$$

For fixed $\ell \geq 1$, divide by $R_t^\ell \geq t^\ell$. The upper bound tends to zero because $t^{\alpha-\ell} \log t \rightarrow 0$ and $\alpha < 1 \leq \ell$. Hence $d_\ell = 0$ for every positive ℓ . This proves that h , and therefore H , is constant. Its value is determined without modifying the original source: $H(0) = F(0)/P(0) = F(0)$. The result is the literal factorization

$$F(z) = F(0) \prod_j (1 - z/a_j). \quad (\text{GFR19})$$

This proof does not invoke the conclusion of a factorization theorem as an input. The counting, convergence, selected-circle lower bound, and Fourier-coefficient bound together prove the entire special case used by the workbench.

5. The logarithmic coefficient map retains every multiplicity

Group repeated zeros and put $\beta_j = 1/a_j$, now indexing distinct zeros with multiplicities m_j . Equation (GFR11) gives $S = \sum_j m_j |\beta_j| < \infty$, and the β_j are bounded because there is a zero-free neighborhood of zero. Choose $r > 0$ with $r|\beta_j| \leq 1/2$ for all j ; if there are no zeros any positive r works. For $|z| \leq r$,

$$\sum_j \left| \frac{m_j \beta_j}{1 - \beta_j z} \right| \leq 2S. \quad (\text{GFR20})$$

The uniform logarithm convergence used to construct P , together with this uniform derivative bound, permits termwise differentiation there. Equivalently, integrate this uniformly convergent derivative series from zero and compare the analytic logarithms, whose values at zero are fixed. In either construction,

$$-\frac{F'(z)}{F(z)} = \sum_j \frac{m_j \beta_j}{1 - \beta_j z}.$$

The double geometric series is absolutely summable because

$$\sum_j \sum_{n \geq 0} m_j |\beta_j|^{n+1} r^n = \sum_j \frac{m_j |\beta_j|}{1 - r |\beta_j|} \leq 2S. \quad (\text{GFR21})$$

Thus its sums can be interchanged, and the coefficient of z^n is exactly $\sum_j m_j \beta_j^{n+1}$. This proves GF13 and the workbench's logarithmic coefficients in their original scalar coordinate. It leaves the reflected packet modules, their nilpotents, and their independent operator constructions unchanged upstream of that scalar map.

6. Verification of the positive tails and the exact omitted regions

For the half-line moment integrand define, exactly as in TT1,

$$x_n(y) = \pi n^2 e^{2y}, \quad f_{J,n}(y) = y^J e^{y/2} (16x_n(y)^2 - 24x_n(y)) e^{-x_n(y)}. \quad (\text{GFR22})$$

Here $y \geq 0$, $n \geq 1$, and J is a nonnegative integer. At $J = 0$, y^J is the constant polynomial one. For even J , summing and integrating (GFR22) is precisely M_J in (GFR3): the source factor two and the reflection factor two are both already present in the coefficients 16 and -24 .

Because $x_n(y) \geq \pi > 3/2$, $16x_n^2 - 24x_n = 8x_n(2x_n - 3) > 0$. Therefore

$$0 \leq f_{J,n}(y) \leq 16\pi^2 n^4 y^J e^{9y/2 - \pi n^2 e^{2y}}. \quad (\text{GFR23})$$

The finite callback keeps $1 \leq n < N$ on $[0, L]$. Up to their measure-zero common boundary, the omitted regions are exactly $n \geq N$, $0 \leq y \leq L$ and $n \geq 1$, $y \geq L$. They are disjoint and their union is the full complement of the retained region. Positivity permits the sum and integral interchanges by Tonelli's theorem; the bounds below prove their total finiteness.

On $[0, L]$, the majorant in (GFR23) is at most $16\pi^2 n^4 L^J e^{9L/2 - \pi n^2}$. Integrating the interval of length L and setting $a_n = n^4 e^{-\pi n^2}$ gives the upper bound $16\pi^2 L^{J+1} e^{9L/2} \sum_{n \geq N} a_n$. For every $n \geq N$,

$$\frac{a_{n+1}}{a_n} = (1 + 1/n)^4 e^{-\pi(2n+1)} \leq e^{4/n - \pi(2n+1)} \leq e^{4/N - 2\pi N} =: \rho_n. \quad (\text{GFR24})$$

The final symbol is constant for the selected first omitted index N ; its subscript does not introduce a summation over ratios. For $\rho_n < 1$, induction and the geometric sum give the full upper bound

$$T_n = \frac{16\pi^2 L^{J+1} e^{9L/2} N^4 e^{-\pi N^2}}{1 - e^{4/N - 2\pi N}}. \quad (\text{GFR25})$$

For the omitted y -interval, apply the same exact ratio at fixed $y \geq 0$:

$$\frac{(n+1)^4 e^{-\pi(n+1)^2 e^{2y}}}{n^4 e^{-\pi n^2 e^{2y}}} \leq e^{4/n - 2\pi n e^{2y}} \leq e^{4 - 2\pi e^{2y}} \leq \rho_y := e^{4 - 2\pi} < 1. \quad (\text{GFR26})$$

The first term is $e^{-\pi e^{2y}}$; hence the full n -sum in (GFR23) is bounded by it divided by $1 - \rho_y$. Put $V_J(y) = y^J e^{9y/2 - \pi e^{2y}}$, which is positive on $[L, \infty)$ when $L > 0$. Its logarithmic derivative is

$$\frac{V_J'(y)}{V_J(y)} = \frac{J}{y} + \frac{9}{2} - 2\pi e^{2y} \leq \frac{J}{L} + \frac{9}{2} - 2\pi e^{2L} = -\delta, \quad \delta = 2\pi e^{2L} - \frac{9}{2} - \frac{J}{L}. \quad (\text{GFR27})$$

When $\delta > 0$, integration from L to y gives $V_J(y) \leq V_J(L) e^{-\delta(y-L)}$. Its integral is at most $V_J(L)/\delta$. Thus the second omitted region is bounded above by

$$T_y = \frac{16\pi^2 L^J e^{9L/2 - \pi e^{2L}}}{(1 - e^{4 - 2\pi})(2\pi e^{2L} - 9/2 - J/L)}. \quad (\text{GFR28})$$

These are exactly TT6 and TT10. At the actual parameters $N = 20$, $L = 4$, $0 \leq J \leq 64$, the elementary bounds $\pi > 3$ and $e^8 > 1 + 8 = 9$ give

$$4/N - 2\pi N < -599/5, \quad 4 - 2\pi < -2, \quad \delta > 54 - 9/2 - 16 = 67/2 > 0. \quad (\text{GFR29})$$

All denominators are strictly positive. Consequently the literal omitted integral belongs to $[0, T_n + T_y]$. The negative summand $-24x_n$ remains part of the exact integral and finite callback; its removal in (GFR23) is solely the proved inequality bounding the omitted tails.

7. Analytic callback and source-to-proof comparison

The retained callback extends to every complex y as

$$y^J e^{y/2} \sum_{n=1}^{N-1} (16\pi^2 n^4 e^{4y} - 24\pi n^2 e^{2y}) e^{-\pi n^2 e^{2y}}. \quad (\text{GFR30})$$

It is a finite sum of products and compositions of entire functions, and y^J is a polynomial. Thus it is entire, including at zero for every allowed J . It has no branch choice or hidden singularity. Treating π as its exact real constant, or any fixed constant value in an enclosing ball, does not alter this analyticity statement. The callback's complex ball value must enclose the exact expression; a possibly wide or nonfinite enclosure is a numerical issue, not an analytic singularity. The retained caller separately rejects a nonfinite integration result.

The separate full code comparison verified the following literal correspondence: the callback loops through $n = 1, \dots, N - 1$; tn is (GFR25); ty and decay are (GFR28) and (GFR27); the default values are $N = 20, L = 4$; dimension 16 requests exactly $J = 0, 2, \dots, 64$; the retained integral is partitioned into the eight half-unit intervals with endpoints $j/2, (j+1)/2, 0 \leq j \leq 7$. The denominator comparisons must succeed before the code proceeds. The code forms `arb(0).union((tn+ty).upper())`, which encloses the proved interval $[0, T_n + T_y]$ whenever the computed balls enclose the stated expressions. Ignoring the callback's analytic flag is consistent with the entire formula (GFR30). This establishes the exact code-to-tail-formula match without asserting an integration replay.

8. Acceptance and scope

Every numbered mathematical step GF1–GF13 and TT1–TT12 is accepted. The original-theta specialization of GF1 has the explicit constants in (GFR4) and (GFR9), and its product and logarithmic coefficients follow from the complete proof above. The original half-line tail formulas have the exact constants and denominator signs in (GFR25)–(GFR29). No alteration of the original input, source mass, coefficient coordinate, multiplicity, finite integration interval, or first omitted index is needed.

This review establishes these analytic statements and their exact callback correspondence. It makes no claim that a finite collection of positive Hankel matrices proves RH or that the broader absolute-base geometric program fails. Those conclusions do not follow from the statements proved here.